The Effects of Corporate Purchase Programmes on Market Power

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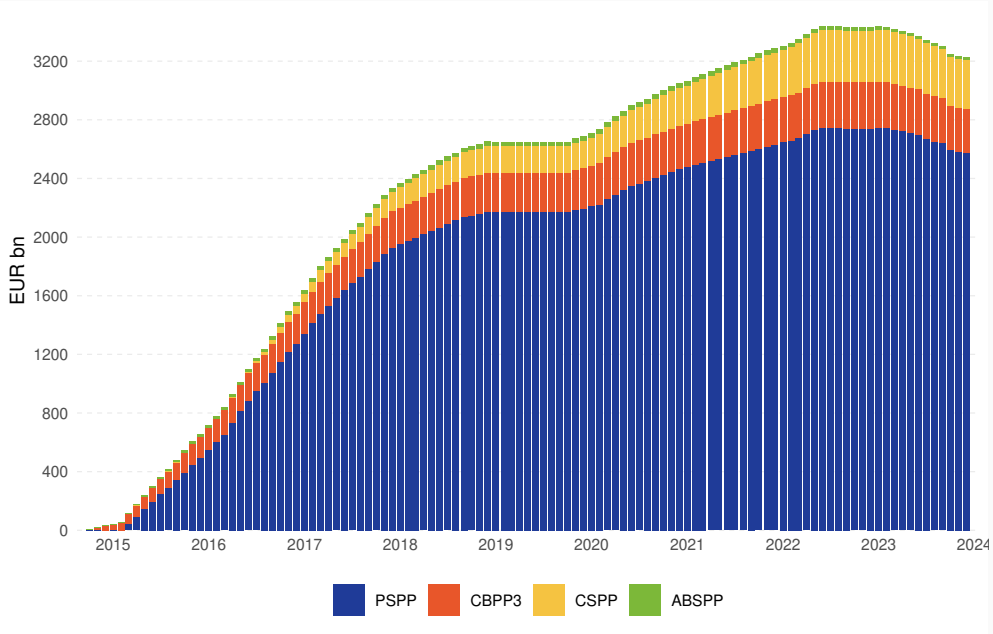
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Introduction

Asset and Corporate Purchase Programmes

- **Central Banks' Mandate:** price stability (low, positive inflation)
- **Problem:** Great Financial Crisis ⇒ rates at Zero Lower Bound
- **Solution:** Asset Purchase Programmes, central bank purchasing financial assets to raise prices and reducing rates
- Mainly purchase of government bonds, but ECB purchased also corporate bonds from firms (CSPP)



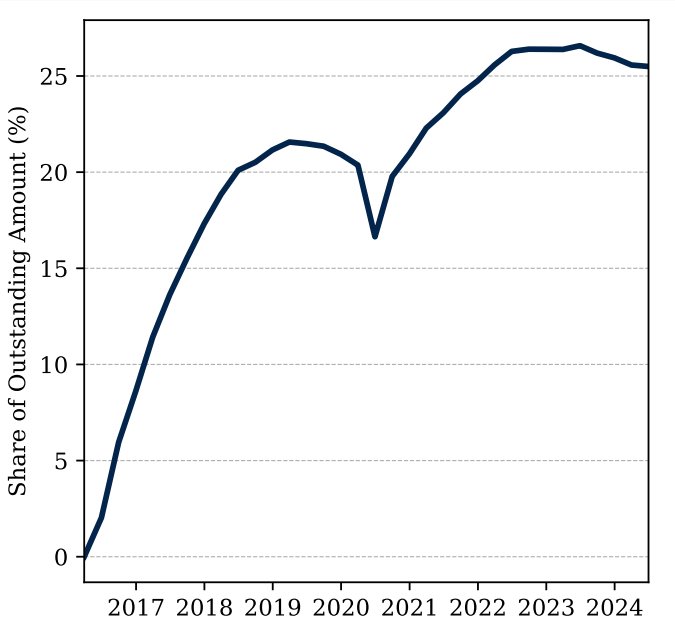


Figure 1: ECB purchases, total and as share of eligible market

Policy Background: Aim of Financial Market Neutrality

As stated by the ECB:

> *In pursuing its objective of maintaining price stability, the ECB is mandated to act in accordance with the principle of an open market economy with free competition [...]. Consequently, the ECB aims for a* **market-neutral implementation** *of the APP [...].*
>
> *— ECB, Economic Bulletin 4/2017*

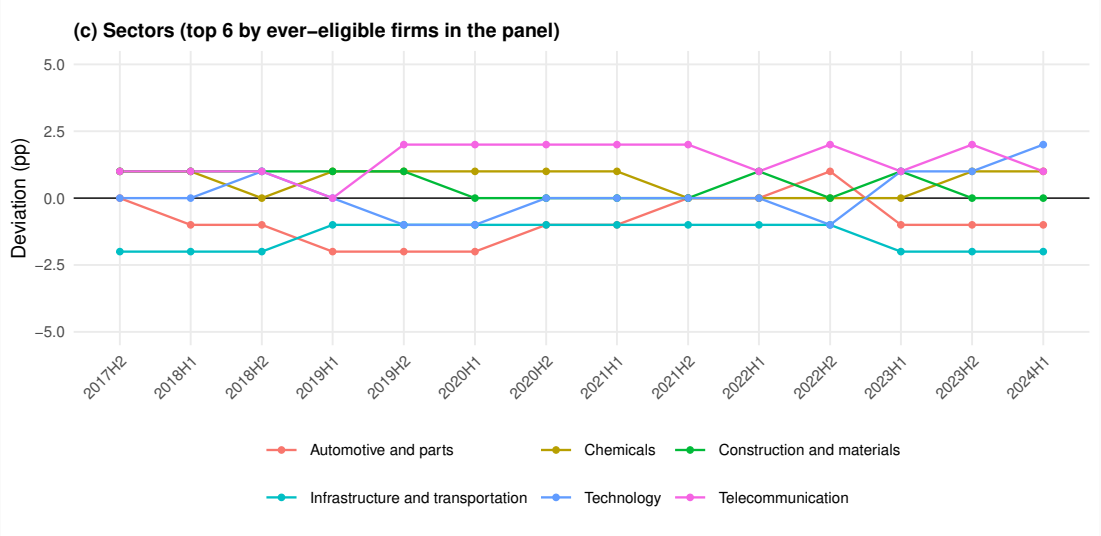
> *In order to [...]* **avoid undue market distortions**, *the Eurosystem does not discriminate on the basis of the issuers' economic activity when determining asset eligibility and actual purchases [...].*
>
> *— Draghi, letter to MEP Urtasun (2018)*

Implementation: internal **benchmark**: purchase across issuer groups in proportion to their eligible outstanding bonds

Policy Background: Potential Sources of Non-Neutralities

Deviations:

Countries and ratings



Eligibility Criteria:

Complete eligibility criteria

- Investment-grade rating ($\geq$ BBB-)

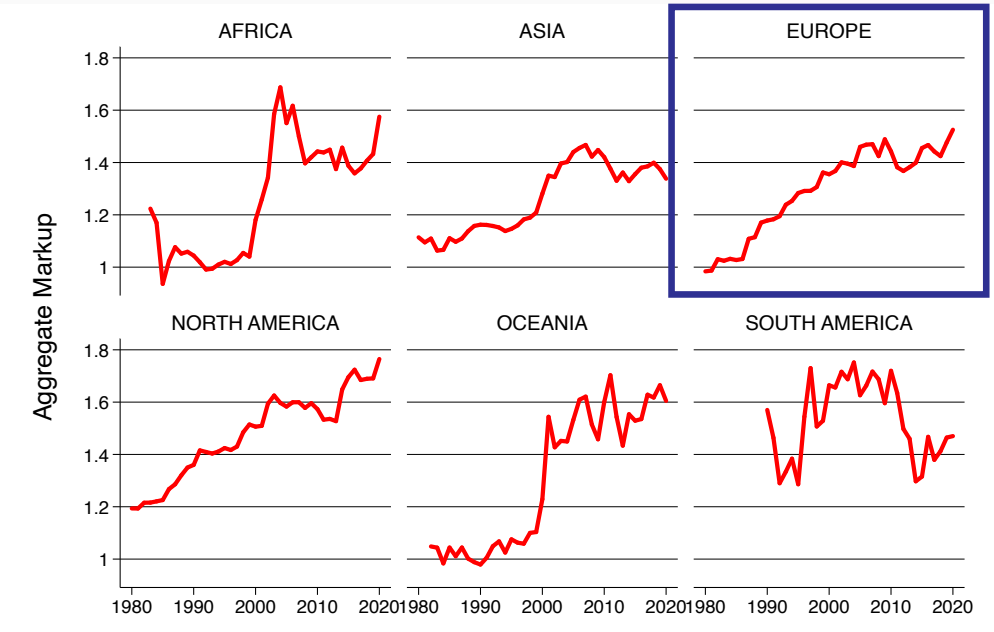
+ Characteristics of Euro-Area Bond Issuers:

- More than 63% of listed groups never issued a bond as of 2018
- **Investment-grade rating:** 15.5% of total bond market; concentration: 39% of Q4 vs less than 3% of other quartiles

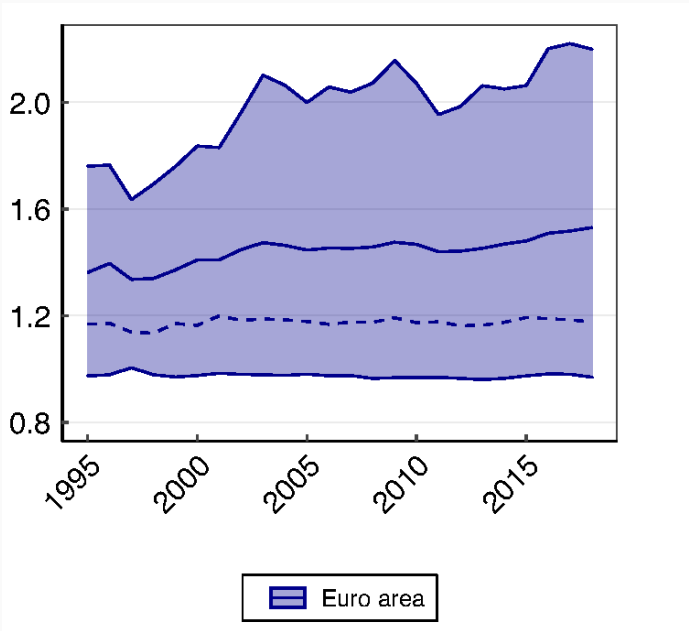
$\Rightarrow$ **Small subset of big corporations** (over half of total revenue)

Market Power Trends

- **Global markups' increase**, Europe from 1.0 in 1980 to 1.52 in 2020
- **Distributional problem:** increase driven by top decile firms
- **Negative impact:** decline in labour and capital share, decrease in long-run interest rate and in startup rates



(a) Aggregate markup by continent (De Loecker and Eeckhout, 2025)



(b) Euro area: mean and 10–90 pctls (Kouvavas et al., 2021)

Figure 2: Markups in Europe: rising average and distribution

Research Question:

Do Corporate Purchase Programmes affect market power by treating firms' input costs unequally through financing?

ECB Assessment:

> *The* **effects** *of our corporate bond purchases on* **competition** *in the Internal Market and* **market concentration** *are expected to have been* **minimal at most** *[. . .].*
>
> — *Lagarde, letter to MEP Van Overtveldt (2021)*

Closest Evidence:

- Acharya et al. (2025): US firms subsidised through QE-induced portfolio rebalancing gained market share
- Adelino et al. (2023): CSPP-eligible suppliers expanded customer base and raised concentration of their industries

Examined Non-Neutralities Effects: Climate (Papoutsi et al., 2021)

This Paper

Empirics:

- CSPP Purchase Effects: purchased vs eligible-not-purchased;
- **Main Result:** 7.1 percent markup increase for purchased firms;
- **Transmission Channel:**
  - Hypothesis: CSPP $\Rightarrow \downarrow$ bond yield $\Rightarrow \uparrow$ capital $\Rightarrow \uparrow$ markup
  - 21.7 b.p. lower yield at issuance;
  - 2.2 p.p. higher investment rate;
  - **Effects on markups and capital coming only from repeated, primary market purchases.**

Theory:

- **Model:** GE with endogenous markups and financial frictions
  - Purchases in segmented financial market $\Rightarrow$ competition effects;
  - Predictions about which purchases can have real effects.
- **Quantification:**
  - Calibrated to measured yield effect; real responses untargeted;
  - Counterfactuals: current design among those with highest welfare.

Related Literature

- **CSPP Effects:** Todorov (2020), Zaghini (2019), De Santis–Zaghini (2021), Galema–Lugo (2021), Adelino et al. (2023)

  My paper: Which purchases have real effects: bond prices to markups

- **Market Power:** De Loecker et al. (2020), De Loecker–Eeckhout (2025), Mongey (2021), Baqaee et al. (2024)

  My paper: Opposite direction: monetary policy $\Rightarrow$ market power

- **Financial Frictions, MP Investment Channel:** Bernanke et al. (1999), Cooley–Quadrini (2001), Ottonello–Winberry (2020), Jeenas (2026)

  My paper: Extend to effects on markups and market power

- **UMP and QE:** Gertler–Karadi (2011), Curdia–Woodford (2011), Papoutsi et al. (2021), Gilchrist et al. (2024), Holm-Hadulla–Leombroni (2025)

  My paper: UMP in an oligopoly economy

- **Oligopoly in Macro:** Atkeson–Burstein (2008), Edmond et al. (2015), De Loecker et al. (2021), Wang–Werning (2022), Doraszelski et al. (2022)

  My paper: Add financing stage in GE

Outline

1. Introduction
2. Empirical Analysis
    - Data Work
    - Identification of Purchase Effects
    - Testing Issuing Conditions
    - Testing Capital Channel
    - Testing Market Power
3. Analytical Model
    - Environment
    - Solution
4. Quantitative Model
    - From Analytical to Quantitative
    - Calibration
    - Validation
    - Comparison of Policy Designs
5. Conclusions

Empirical Analysis

Treatment Identification

Purchased bonds:

- ECB publishes its portfolio of securities weekly (no amounts info)
- Each portfolio entry is classified either as:
  - **Issuance-week purchases:** bonds purchased within a week from issuance
  - Secondary market purchases: bonds purchased more than a week after issuance
  - Benchmark portfolio bonds: bonds published in the first publication on 23 June 2017

Eligible bonds:

- ECB publishes the eligible assets for any programme daily
- Find assets that satisfy the eligibility criteria of the CSPP

Dataset Construction

Steps:

- Download information on all Euro-denominated bonds issued by non financial issuers between 2010 and 2024
- Manually link every asset to the main parent of the issuing company
- Construct measure of bonds issued and bonds outstanding in each year for every issuer
- Reconstruct the CSPP benchmark: issuer group's share of all eligible outstanding bonds $\Rightarrow$ control for the purchase rule
- Download financial information of firms for panel analysis

Results:

- Panel dataset at firm-year level Summary statistics
- Cross-sectional dataset at bond level Summary statistics

Some Facts about the CSPP Data

By year

	Issues	Purchased	Share
Issuance-week purchase probability, by the issuer's previous issue			
First eligible issue of the issuer	340	195	**0.574**
Previous issue purchased in its issuance week, any gap	957	727	0.760
within 1 year	762	627	**0.823**
within 90 days	495	447	0.903
Previous issue not purchased in its issuance week, any gap	955	161	0.169
within 1 year	843	126	**0.149**
within 90 days	531	55	0.104
Last two issues both purchased in their issuance week	637	493	0.774

Identification of Purchase Effects

Focus: Purchase Effects – purchased vs eligible-not-purchased

Concern 1: Level Selection – *who* the ECB buys

Firms: *who*, not *when*

	Across firms	Within firm
	Country, Sector×Year FE	Firm, Year FE
Benchmark share	12.8879***	1.0910
	(2.4897)	(6.0225)
Cluster-robust F	26.80	0.03
Observations	1289	1372

Bonds: purchase within yield cells

Within-R^2 of purchase on:	
Amount, duration	0.0029
Amount, duration, coupon	0.0076
Benchmark share, amount, duration	0.0038
Purchased − non-purchased, within cells:	
Log amount issued	0.108**
	(0.053)
Log duration	-0.004
	(0.040)
Observations	2,920

$\Rightarrow$ Rule explains *who*, not *when* $\Rightarrow$ **within-firm variation**

- Design: eligible units only, benchmark control, long differences

Identification of Purchase Effects

Focus: Purchase Effects – purchased vs eligible-not-purchased

Concern 1: Level Selection – *who* the ECB buys

$\Rightarrow$ Rule explains *who*, not *when* $\Rightarrow$ **within-firm variation**

Concern 2: Timing Selection – *when* the ECB buys

	Across firms	Within firm
	Country, Sector×Year FE	Firm, Year FE
Benchmark share	8.4292***	5.9164
	(3.1409)	(6.6361)
Log assets	0.0652***	-0.0193
	(0.0187)	(0.0490)
Bonds outstanding / assets	0.3971**	-0.6752***
	(0.1976)	(0.2281)
Log markup	0.0766	0.1091
	(0.0615)	(0.1493)
Investment rate	0.4001	0.2892
	(0.6413)	(0.8545)
Within R^2	0.0860	0.0197
Observations	1149	1219

- **Past outcomes** (shaded) do not predict purchase
- **Past characteristics** have low predictive power: within R^2 8.6% across firms, 2.0% within firm

All characteristics

$\Rightarrow$ Timing unrelated to past outcomes $\Rightarrow$ **remaining variation usable**

Testing Issuing Conditions: Regression Details

Analysis at the bond (b) level:

$$y_b = \beta D_b + \delta X_b + \alpha_{fe} + \varepsilon_b$$

- **Outcome variable:** yield to maturity (YTM) at issuance
- **Treatment D:** 1 if bond b purchased by ECB
- **Identification:** within fixed-effect cells (conditional independence); identified biases run towards zero. Details
- Fixed Effects α_{fe}: issuer, rating, sector, country, **eligible × year**
- Controls X_b: CSPP benchmark, maturity, amount issued
- SE clustered at issuer level; outliers trimmed at 1% level

Testing Issuing Conditions: Results

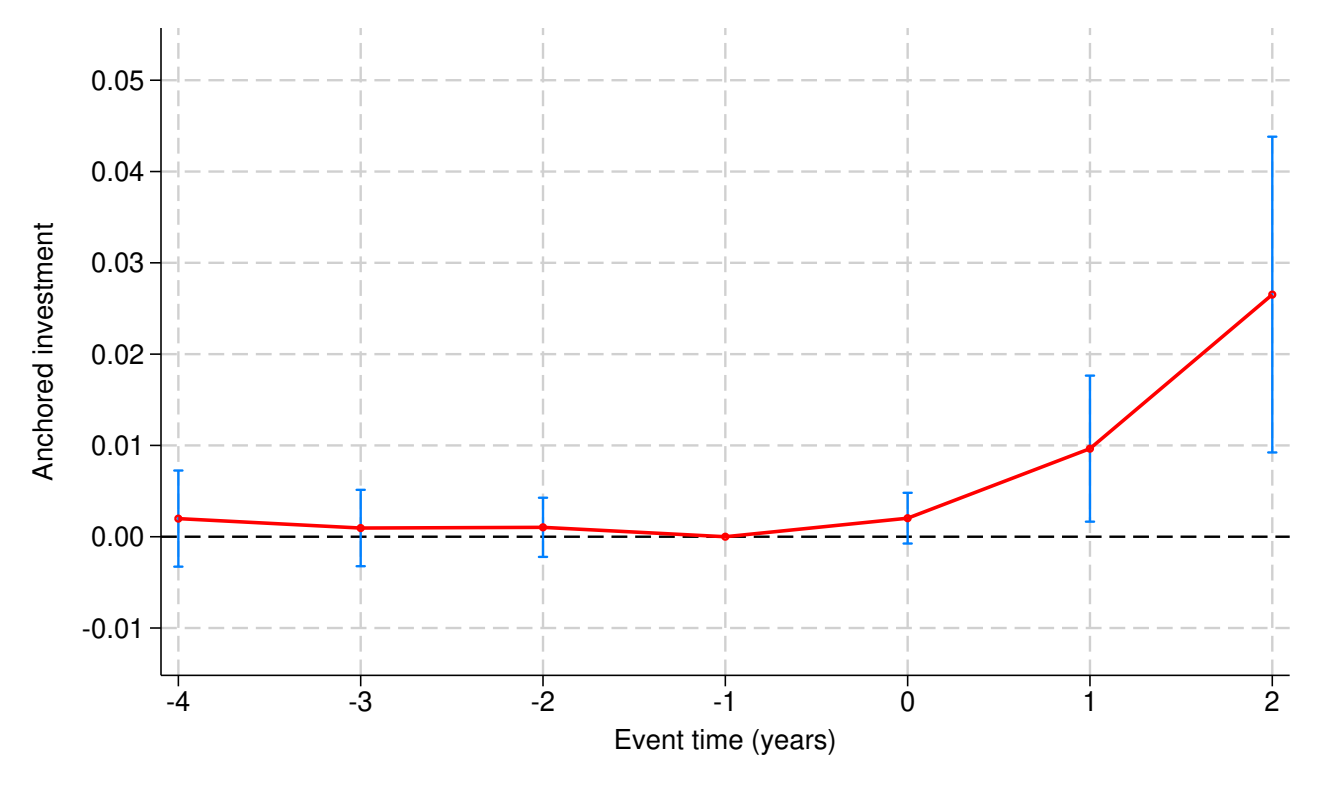
Yield to Maturity	(1)	(2)	(3)	(4)	(5)
Treated	**-0.2170*****	-0.1948**	-0.2484**	-0.2427*	-0.1694*
	(0.0809)	(0.0901)	(0.1161)	(0.1326)	(0.1028)
Controls	Yes	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓	
Issuer × Year FE					✓
NACE × Year FE	✓	✓			
Country × Year FE	✓	✓			
NACE × Country × Year FE			✓	✓	
Eligible × Year FE	✓	✓	✓	✓	✓
Bond Rating FE	✓		✓		✓
Rating × Year FE		✓		✓	
Observations	2,920	2,873	2,405	2,361	2,217
R^2	0.9075	0.9146	0.9267	0.9336	0.9379
Adjusted R^2	0.8564	0.8587	0.8690	0.8716	0.8989

Testing Capital Channel: Regression Details

$$y_{i,t+h} - y_{i,t-1} = \beta_h^{LP-DiD} \Delta D_{i,t} + \delta_h X_{i,t-1}^h + \alpha_t^h + \alpha_c^h + \alpha_{s,t}^h + \varepsilon_{i,t}^h$$

- **Outcome variable:** Cumulative investment rate: $\frac{\text{capex}_{i,t+h} - \text{capex}_{i,t-1}}{\text{assets}_{i,t-1}}$
- **Treatment D:** 1 if firm i has at least one issuance-week purchased bond by ECB in year t
- **Identification:** Sample restricted to ever-eligible issuers
- **Estimator:** LP-DiD, Non-Absorbing version $\Rightarrow \beta_h$ identified from treatment switches vs clean controls; Estimator
- Controls $X_{i,t-1}$: CSPP benchmark, log assets, (as ratio over assets) bonds outstanding, total liabilities, total debt, cash, gross profits, wage bill, selling-general-administrative
- FEs: time + country + sector $\times$ time
- SE clustered at firm level; outliers trimmed at 1% level

Testing Capital Channel: Results



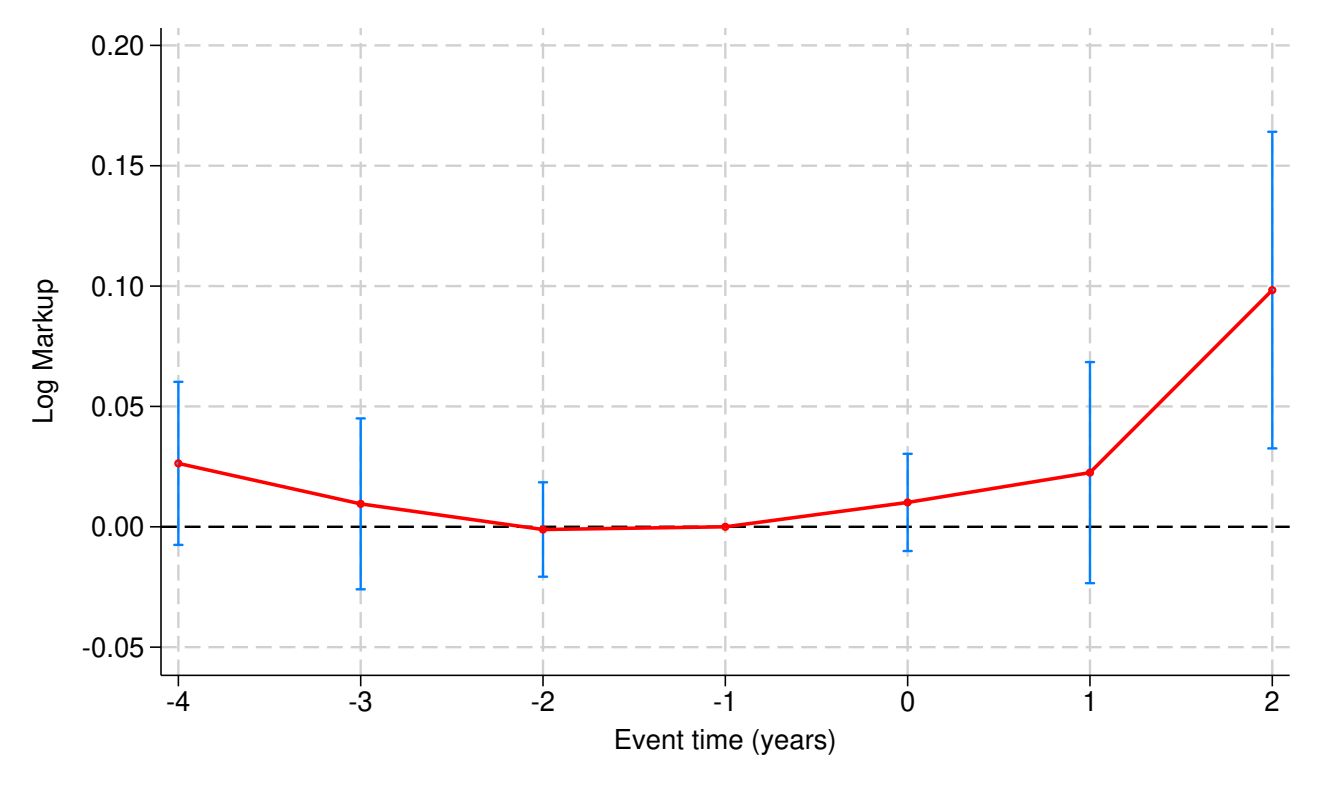
Pooled Estimates						
	Estimate	SE	p-value	90% CI		Obs
Pre	0.0004267	0.0028454	0.8810	-0.0042887	0.0051422	536
Post	**0.0215846**	0.0092816	0.0216	0.0062105	0.0369586	599

Testing Market Power: Regression Details

$$y_{i,t+h} - y_{i,t-1} = \beta_h^{LP-DiD} \Delta D_{i,t} + \delta_h X_{i,t-1}^h + \alpha_t^h + \alpha_c^h + \alpha_{s,t}^h + \varepsilon_{i,t}^h$$

- **Outcome variable:** (log) Markup Estimation Details
- **Treatment D:** 1 if firm i has at least one issuance-week purchased bond by ECB in year t
- Controls $X_{i,t-1}$: CSPP benchmark, log assets, (as ratio over assets) bonds outstanding, total liabilities, total debt, cash, gross profits, wage bill, selling-general-administrative
- FEs: time + country + sector $\times$ time
- SE clustered at firm level; outliers trimmed at 1% level

Testing Market Power: Results



Pooled Estimates

	Estimate	SE	p-value	90% CI		Obs
Pre	0.0132584	0.0142507	0.3540	-0.0103627	0.0368796	536
Post	**0.0692102**	0.0298974	0.0222	0.0196749	0.1187454	592

Which Purchases Have Real Effects

Only issuance-week purchases matter

Pooled post effect	Investment rate	Log markup
Issuance-week purchase (headline)	0.022^{**}	0.069^{**}
	(0.009)	(0.030)
dropping firms also bought on the secondary market	0.018	0.139^{***}
	(0.012)	(0.044)
controlling for secondary-market purchases	0.021^{**}	0.061^{*}
	(0.009)	(0.036)
Secondary-market purchase only	0.011	-0.044
	(0.009)	(0.053)

$\Rightarrow$ Effect only when **funds go directly to the issuing firm**

Which Purchases Have Real Effects

Only issuance-week purchases matter

$\Rightarrow$ Effect only when **funds go directly to the issuing firm**

Only repeated purchases matter

Pooled post effect	Investment rate	Log markup
Continuous purchases (3 consecutive years)	0.0216^{**}	0.069^{**}
	(0.0093)	(0.030)
One-shot purchase (single purchase year)	-0.0026	-0.005
	(0.0030)	(0.026)
Treated events: continuous / one-shot	12 / 53	11 / 51

$\Rightarrow$ **Single purchase has no effect**

Which Purchases Have Real Effects

Only issuance-week purchases matter

$\Rightarrow$ Effect only when **funds go directly to the issuing firm**

Only repeated purchases matter

$\Rightarrow$ **Single purchase has no effect**

No announcement effect

	Investment rate	Log markup
Eligible at the announcement (March 2016)	−0.0004	−0.010
	[−0.0071, 0.0062]	[−0.047, 0.027]
Observations	3,476	3,920
Firms	496	624

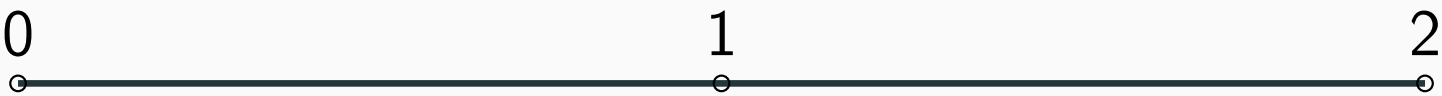
- Eligible vs never-eligible issuers around 2016; firm, country × year, size × year FE

Analytical Model

Model — General Setup

- **3-period GE model**
- **Agents:** household, final good producer, intermediate producers, private financial intermediary, central bank
- Continuum of markets indexed by j, each with I firms that compete à la Cournot $\Rightarrow$ **Endogenous markups**
- Firms' **financing constraint:** issue bonds to install capital
- **Financial frictions:** convex holding costs $\Rightarrow$ downward sloping demand curve for bonds
- No uncertainty; Policy introduced as unexpected (MIT) shock

Model — Timing



0) Central Bank announces $\bar{\phi}$

1) Firms issue $a_{t,j}(i)$

2) Central Bank buys $a^{CB}_{t,j}(i)$, purchase path realises

3) Private bank clears market $a^{B}_{t,j}(i)$ and pins down price $Q_{t,j}(i)$

1) First production $y_{t+1,j}(i)$

2) Firms can issue $a_{t+1,j}(i)$

3) Central Bank buys revealed $a^{CB}_{t+1,j}(i)$

4) Private bank clears market $a^{B}_{t+1,j}(i)$ and pins down price $Q_{t+1,j}(i)$

1) Second production $y_{t+2,j}(i)$

Model — Environment: Household and Final Producer

Agg Constr

Household:

$$\max_{C_t, N_t, S_t} \sum_{t=0}^{2} \beta^t \left(C_t - \frac{N_t^{1+\frac{1}{\varphi}}}{1+\frac{1}{\varphi}} \right)$$

$$\text{s.t.} \quad C_0 + Q_0^s S_0 = E \qquad \text{for } t = 0$$

$$P_t C_t + Q_t^s S_t = W_t N_t + S_{t-1} + D_t + T_t \qquad \text{for } t = 1, 2$$

$$\Rightarrow \text{EE: } Q_t^s = \beta \mathbb{E}_t \left[\frac{P_t}{P_{t+1}} \right] \qquad \text{LS: } N_t^{\frac{1}{\varphi}} = \frac{W_t}{P_t}$$

Final good producer:

$$Y_t = \left[\int_0^1 Y_{t,j}^{\frac{\theta-1}{\theta}} \, dj \right]^{\frac{\theta}{\theta-1}} \qquad Y_{t,j} = \left[I^{-\frac{1}{\eta}} \sum_{i=1}^{I} y_{t,j}(i)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}$$

$$\Rightarrow p_t(i) = I^{-\frac{1}{\eta}} \, y_{t,j}(i)^{-\frac{1}{\eta}} \, Y_{t,j}^{\frac{1}{\eta}-\frac{1}{\theta}} \, Y_t^{\frac{1}{\theta}} \, P_t$$

Model — Environment: Intermediate Firms (1)

Intermediate goods producers:

- Finite number I of firms in each j sector, which compete oligopolistically setting quantities

- **Production function:**

$$y_{t+1,j}(i) = z_{t+1,j}(i) k_{t,j}(i)^{\alpha} n_{t+1,j}(i)^{1-\alpha} \tag{1}$$

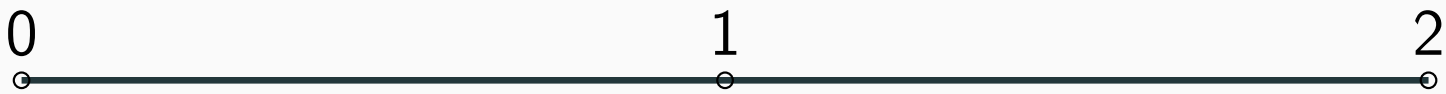
- **Financing constraint** in $t = 0$:

$$k_{t,j}(i) = e_j(i) + Q_{t,j}(i) a_{t,j}(i) \tag{2}$$

Model — Environment: Intermediate Firms (2)

Full Problem

Intermediate goods producers - Timing:



1) Issue a_0

$\Rightarrow k_0 = e + Q_0 a_0$

1) Choose n_1 to max Υ_1

2) Repay $(1-\lambda)a_0$

3) Install k_1: retained earnings and/or issue a_1;

1) Choose n_2 to max Υ_2

2) Repay remaining debt position

3) Pay out div_2

Intermediate goods producers - Labour input choice

$$\Upsilon_1(z_1, k_0, y_1(-i)) \equiv \max_{n_1} \left[p_1 y_1 - W_1 n_1 - W_1 \psi_1 z_1^{\psi_2}\right]$$

$$\text{s.t.} \quad y_1 = z_1 k_0^{\alpha} n_1^{1-\alpha}$$

$$p_1 = I^{-\frac{1}{\eta}} \, y_1^{-\frac{1}{\eta}} \, Y_{1,j}^{\frac{1}{\eta}-\frac{1}{\theta}} \, Y_1^{\frac{1}{\theta}} \, P_1$$

Model — Environment: Private Financial Intermediary

Private financial intermediary:

- Perfectly competitive, owned by the household
- **Financial frictions** as holding costs:

$$h_0^B = \int_j \sum_{i=1}^{I} \frac{\gamma}{2} \frac{\left(Q_{0,j}(i)\, a_{0,j}^B(i)\right)^2}{e_j(i)}\, dj, \tag{3}$$

$$h_1^B = \int_j \sum_{i=1}^{I} \frac{\gamma}{2} \left[\frac{\left(Q_{1,j}(i)\, a_{1,j}^B(i)\right)^2}{k_{0,j}(i)} + \frac{\left(\lambda\, Q_{0,j}(i)\, a_{0,j}^B(i)\right)^2}{k_{0,j}(i)} \right] dj, \tag{4}$$

where:

- a_0^B and a_1^B are the shares of the debt position of firm i, j at time t owned by the bank
- $Q_{t,j}(i)$ are the relative prices of such bonds
- γ: strength of financial frictions in the economy

Model — Environment: Central Bank

Central bank:

- **Policy funding:**

$$Q_t^d \, d_t = \int_j \sum_{i=1}^{I} Q_{t,j}(i) \, a_{t,j}^{CB}(i) \, dj \;\equiv\; V_t^{CB} \tag{5}$$

- **Holding costs:**

$$h_0^{CB} = \int_j \sum_i \frac{\gamma + \kappa}{2} \frac{\left(Q_{0,j}(i) a_{0,j}^{CB}(i) \right)^2}{e_j(i)} \, dj$$

  where κ is an efficiency cost

- **Purchases:** described by the collection $\left(\widetilde{\mathcal{I}}, \; \bar{\phi}, \; \xi_0(\cdot), \; \rho \right)$ with

$$a_t^{CB}(i) = \phi_t(i) \, a_t(i),$$

$$\phi_0(i) = \begin{cases} \bar{\phi} + \xi_0(i) & \text{if } i \in \widetilde{\mathcal{I}}, \\ 0 & \text{otherwise,} \end{cases} \qquad \phi_1(i) = \rho \, \phi_0(i), \quad \rho \in [0, 1].$$

Model — Solution: Intermediate Firms

t=2: Production decision (endogenous markups, static choice)

$$p_2 = \mu_2 \frac{W_2}{z_2 k_1^{\alpha}(1-\alpha) n_2^{-\alpha}} \tag{6}$$

t=1 and t=0: Two endogenous regimes

- *Unconstrained firms* ($div_1 > 0$): $Q_t = M_{t+1} \Rightarrow r_t =$ frictionless
- *Constrained firms* ($div_1 = 0$): Q_t pinned by bank's FOC $\Rightarrow$ higher r_t embedding financial friction

Capital FOC:

$$\frac{\partial \Upsilon_t}{\partial k_{t-1}} = r_{t-1} + \delta$$

Full derivations Bond Mapping

Model — Solution: Private Financial Intermediary

Full bank problem

Problem at t=1:

$$\max_{a_1^B, d_1} \quad V_1^B = \int_j \sum_{i=1}^{I} \left(M_2 - Q_1\right) a_1^B \, dj + \left(M_2 - Q_1^d\right) d_1 - M_2 h_1^B$$

Risk-free debt FOC:

$$Q_t^d = M_{t+1} = \beta$$

Bond pricing FOC:

$$Q_1 = \underbrace{M_2}_{\text{frictionless}} \underbrace{- M_2 \gamma \, Q_1 \, \frac{Q_1 \, a_1^B}{k_0}}_{\text{friction wedge}}$$

$$Q_0 = \underbrace{(1-\lambda) M_1 + \lambda M_1 M_2}_{\text{frictionless}} \underbrace{- M_1 \, Q_0 \left[\gamma \, \frac{Q_0 a_0^B}{e} + M_2 \lambda \gamma \, \frac{\lambda Q_0 a_0^B}{k_0} \right]}_{\text{friction wedge}}$$

Model — Solution: Introduce Policy

Security market clearing - No Policy: (firm by firm)

$$a^B_{t,j}(i) = a_{t,j}(i)$$

Security market clearing - Policy: (firm by firm)

$$a^B_{t,j}(i) = a_{t,j}(i) - a^{CB}_{t,j}(i) = a_{t,j}(i)\Big(1 - \phi_{t,j}(i)\Big)$$

$\Rightarrow$ **Lower effective supply of bonds issued by purchased firms**

$\Rightarrow$ Higher bond price $Q_{t,j}(i)$

$\Rightarrow$ Higher capital $k_{t,j}(i)$, higher output $y_{t+1,j}(i)$, higher markups $\mu_{t+1,j}(i)$

What this Model Can and Cannot Replicate

- **All signs of transmission channel:** Purchase $\Rightarrow \downarrow$ Bond Yield $\Rightarrow \uparrow$ capital $\Rightarrow \uparrow$ markup

- **No effect of secondary market purchases:** If security already cleared the market, purchase changes neither sunk quantity nor realised price $\Rightarrow$ no change in firm's installed capital $\Rightarrow$ production block unchanged;

- **Only repeated purchases have significant effects:** first purchase is only price revaluation of sunk issuance $\Rightarrow$ small effect; second purchase comes with re-optimisation $\Rightarrow$ bigger effect;

- **Capital response from dividend-paying firms:** Purchase of unconstrained firms has no effect on real variable. Contrast with emprirical evidence

Quantitative Model

From Analytical to Quantitative

- **Payout adjustment costs:** marginal value of internal funds varies continuously across firms

$$\varphi(div_1) \;=\; div_1 + \frac{\kappa_d}{k_0}\big(div_1 - \bar{d}(i)\big)^2 \quad \Rightarrow \quad div_1 \;=\; \bar{d} + \frac{k_0}{2\kappa_d}\left(\frac{1}{\vartheta_1} - 1\right)$$

- **Tax advantage of debt:** flat corporate tax τ on operating profit, with deductible interests and depreciation; $t = 1$ budget constraint:

$$\varphi(div_1) \;\leq\; (1-\tau)\Upsilon_1 - a_0\big(1-\lambda-\tau Q_0 r_0\big) + \big(1-(1-\tau)\delta\big)k_0 - k_1 + Q_1 a_1$$

- **Reference holdings and per-security slopes:**

$$h_0^B = \int_j \sum_i \frac{\gamma_0}{2}\, e(i)\Big(\big[x_0(i) - \bar{x}_0\big]^+\Big)^2 dj \qquad \text{with} \quad x_0(i) \equiv \frac{Q_0(i)a_0^B(i)}{e(i)}$$

$$h_1^B = \int_j \sum_i \left[\frac{\gamma_1}{2}\, k_0(i)\Big(\big[x_1(i) - \bar{x}_1\big]^+\Big)^2 + \frac{\gamma_0}{2}\,\frac{\Big(\big[\lambda Q_0(i)a_0^B(i) - \lambda\bar{x}_0 e(i)\big]^+\Big)^2}{k_0(i)}\right] dj$$

Regimes

Calibration - Externally Set Parameters

Parameter	Value	Description	Source
Programme, measured			
$\bar{\phi}$	0.0853	Announced share of eligible issuance bought	Data
ϕ_0	0.131	Per-issue intensity on purchased firms	Data
ρ	0.823	Repurchase probability	Data
	0.574	Purchased share of eligible firms	Data
	0.649	Purchased share of eligible issuance	Data
	0	Eligible share of firms, smallest size quartile	Data
	0.057	Eligible share of firms, second size quartile	Data
	0.297	Eligible share of firms, third size quartile	Data
	0.583	Eligible share of firms, largest size quartile	Data

Calibration - Internally Calibrated Parameters

Fitted in every solve

Parameter	Value	Description	Moment	Model	Data
A. Calibrated on moments					
γ_0	0.00796	Slope of the holding cost, first issue	Empirical result: Yield effect of a purchase (pp)	-0.217	-0.217
$\bar{x}_0$	0.3306	Reference holding, first issue	Data: Bond spread over the AAA euro-area government curve (bp)	119.9	120.0
γ_1	0.065	Slope of the holding cost, repeat issue	Empirical result: Yield effect of a purchase (pp)	-0.191	-0.217
$\bar{x}_1$	0.46	Reference holding, repeat issue	Data: Bond spread over the AAA euro-area government curve (bp)	44.7	120.0
κ_d	0.1226	Payout-adjustment cost	Payout sensitivity to operating profit	0.0545	0.0545

Parameters explored Other safe rates

Validation - Untargeted Moments

		Data	
Moment	Model	Estimate	90% interval
Responses of purchased firms (purchased minus never purchased)			
Capital, $\Delta \log k_1$	0.0310	0.099	[0.028, 0.171]
Payout, $\Delta(div_1/k_0)$, on pre-purchase capital	0.00546	−0.0121	[−0.0298, 0.0055]
Markup, $\Delta \log \mu_2$	0.00205	0.069	[0.020, 0.119]
Single purchase effect on capital ($\Delta \log k_1$)	0.0009	−0.0122	[−0.0352, 0.0107]

Comparison of Policy Designs

The following 9 purchase designs are compared, with the same euros at the first issue:

- Headline programme: main results
- Announced policy (equal shares): no deviations from announcement
- Mid-size firms: target smallest eligible firms (size Q2 and Q3)
- Large firms: target all eligible firms in Q4 of size distribution
- 10% largest firms: target 10% largest firms in Q4
- Low dividend payers: proxy for targeting most constrained firms
- Wider eligibility (headline purchase rule): increase number of eligible firms in Q1 and Q2, purchase as in headline
- Wider eligibility (announced policy): no deviations
- Wider eligibility (smallest firms first): purchase the smallest eligible firms first, up to 70% per issue

Comparison of Policy Designs - Results: Aggregate Output

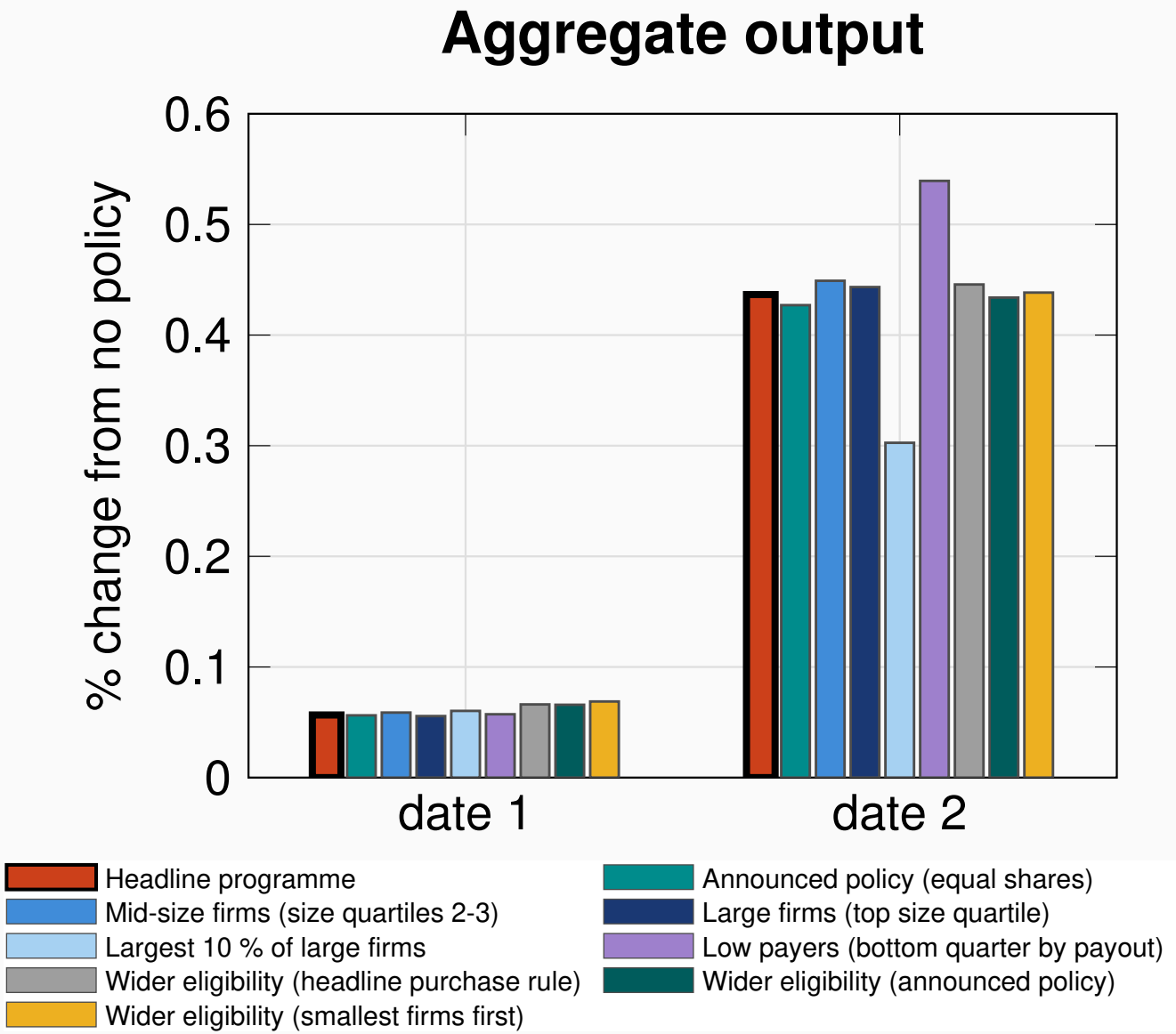
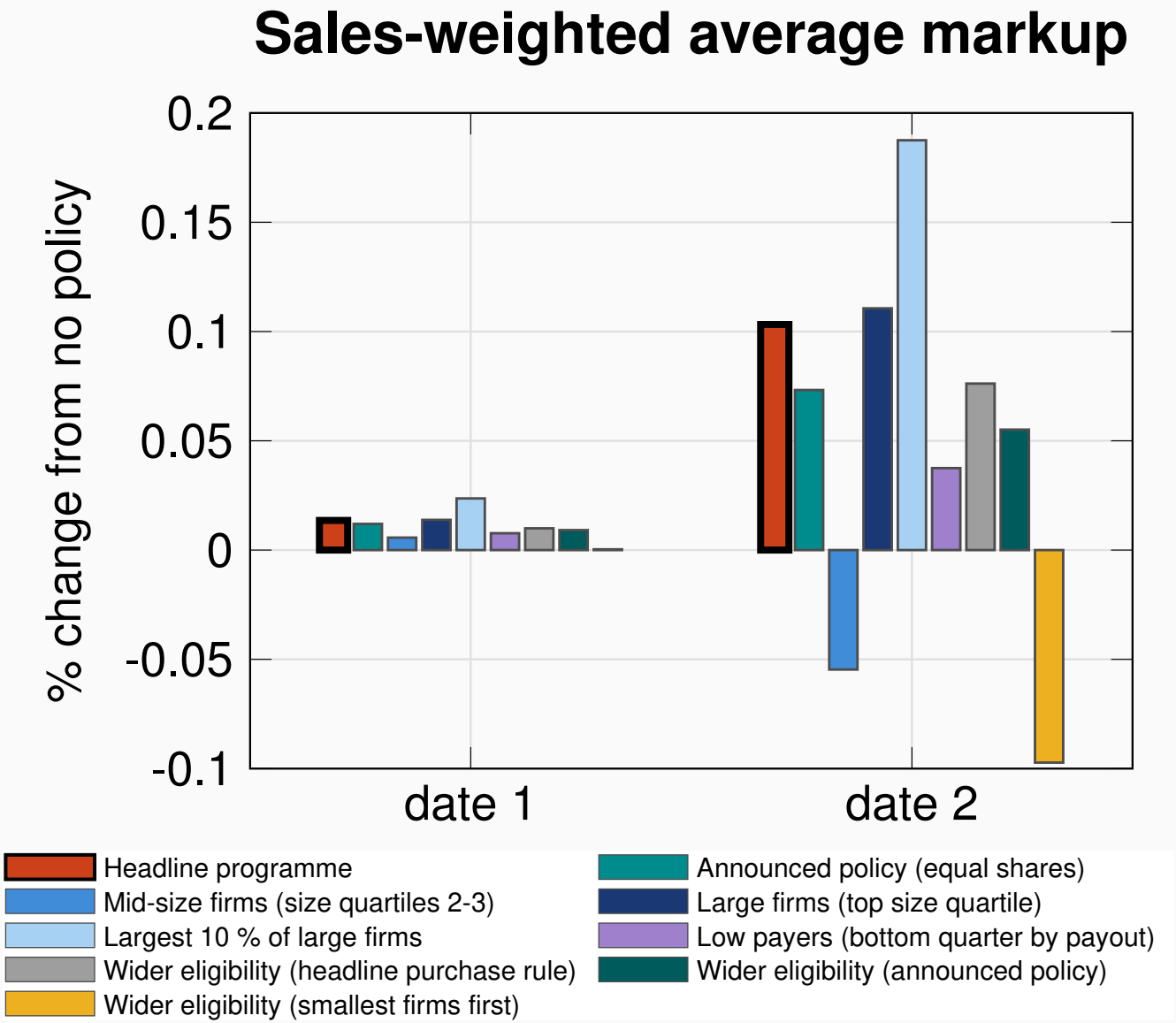


Table More figures

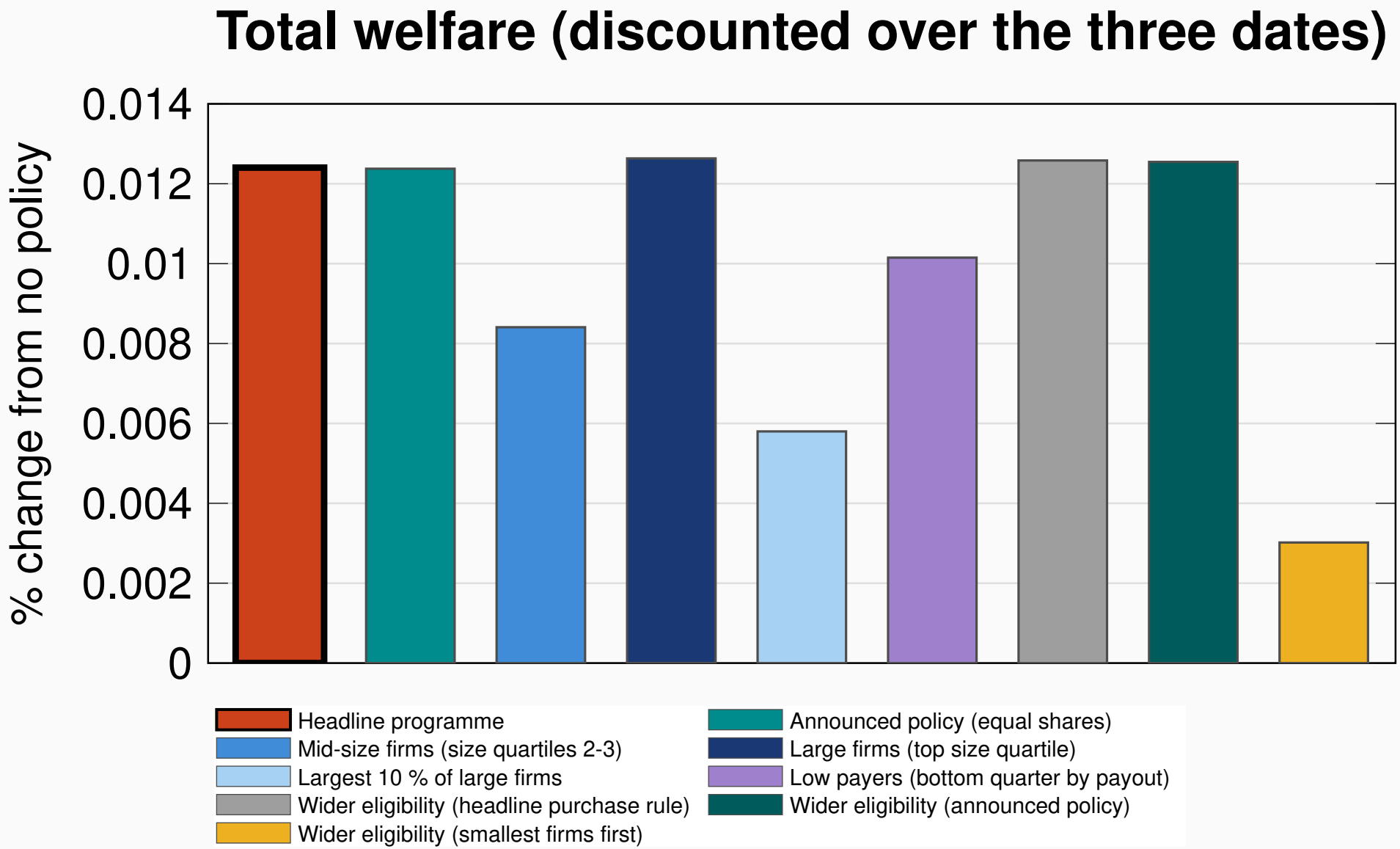
Comparison of Policy Designs - Results: Aggregate Markup



Markups by group Distribution

Comparison of Policy Designs - Results: Welfare

Welfare decomposition



Conclusions

Conclusions

Empirical analysis:

- Evidence of **markup and capital increase** for firms treated under the CSPP
- Evidence of **lower yield to maturity at issuance** for bonds purchased under CSPP

Theoretical model:

- **Policy reduces cost of capital** for purchased firms $\Rightarrow$ increase production and **gain market power**
- **Current policy** design is among those delivering **highest welfare**

Appendix

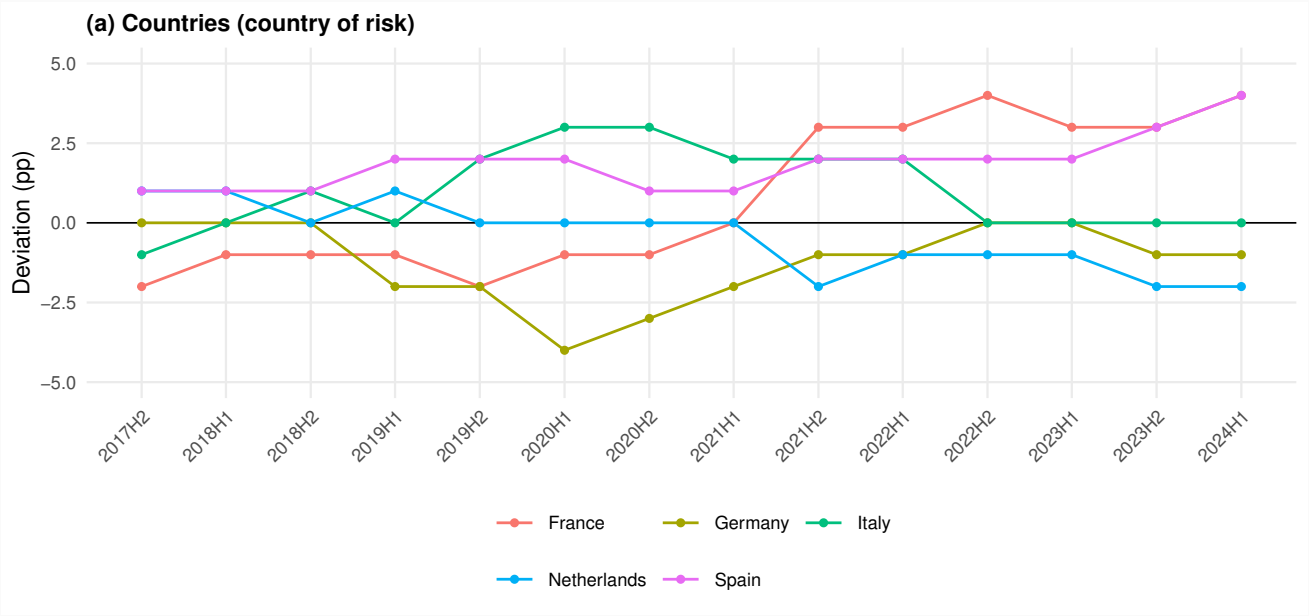
Eligibility Criteria of the CSPP

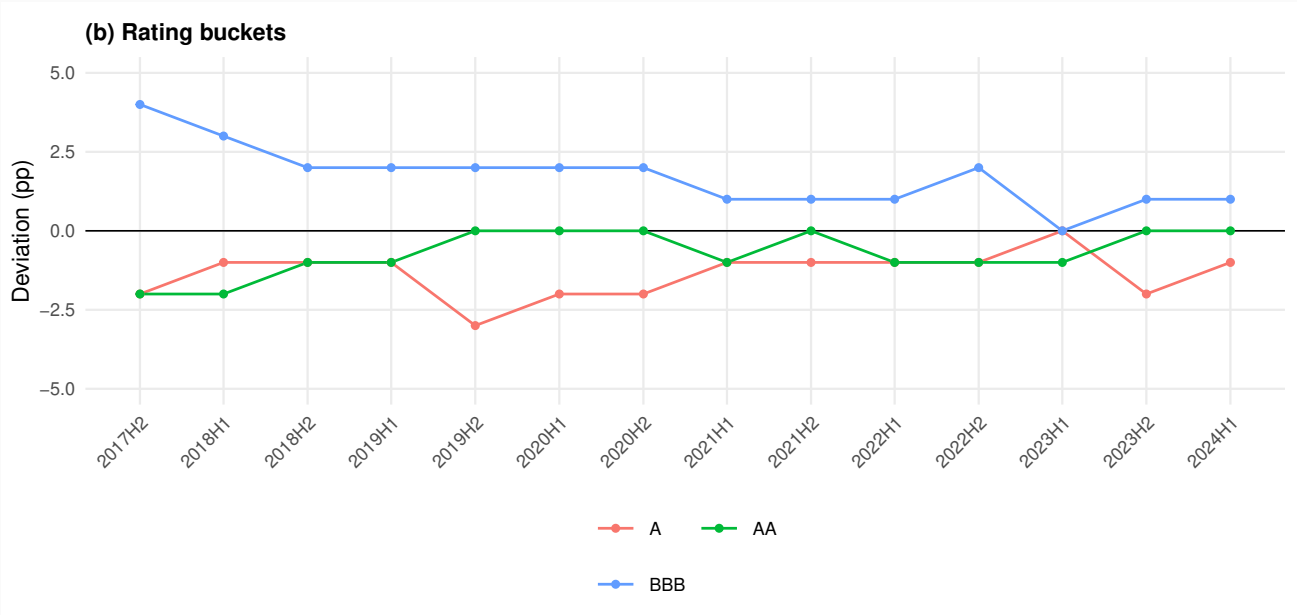
Back

- the bond must be eligible as collateral for Eurosystem credit operations;
- the bond must be denominated in euro;
- the bond must have a minimum first-best credit assessment of at least BBB- or equivalent (obtained from an external credit assessment institution);
- the bond must have a minimum (remaining) maturity of six months and a maximum (remaining) maturity of less than 31 years;
- the issuer must be a corporation established in the euro area, defined as the location of incorporation of the issuer;
- the issuer must not be a credit institution nor have any parent undertaking which is a credit institution.

Deviations from Benchmark: Countries and Ratings

Back





Firm Panel: Summary Statistics (Markup Sample)

Back

Capex sample

	Estimation sample		Mean by group	
	Mean	SD	Ever-tr.	Never-tr.
Markup estimation sample (592 firm-years, 129 firms, 96 ever-treated)				
Markup (level)	1.479	0.614	1.515	1.422
Log markup	0.323	0.354	0.349	0.282
CSPP benchmark (lagged)	0.004	0.011	0.005	0.002
Log assets (lagged)	23.356	1.254	23.543	23.058
Bonds outstanding / assets (lagged)	0.077	0.093	0.065	0.097
Total debt / assets (lagged)	0.284	0.142	0.285	0.282
Total liabilities / assets (lagged)	0.661	0.148	0.638	0.697
Cash / assets (lagged)	0.081	0.062	0.075	0.092
Gross profit / assets (lagged)	0.282	0.152	0.285	0.279
Labour cost / assets (lagged)	0.152	0.100	0.138	0.175
SG&A / assets (lagged)	0.177	0.109	0.166	0.194
Issuance dummy	0.557	0.497	0.582	0.518
Eligible-issuance dummy	0.257	0.437	0.286	0.211
Pays dividends	0.971	0.167	0.988	0.944

Firm Panel: Summary Statistics (Capex Sample)

Back

	Estimation sample		Mean by group	
	Mean	SD	Ever-tr.	Never-tr.
Anchored-capex estimation sample (599 firm-years, 134 firms, 101 ever-treated)				
Capex / lagged assets (level)	0.044	0.029	0.043	0.045
Anchored capex change (regression outcome, pooled τ_0–τ_2)	0.005	0.015	0.005	0.004
CSPP benchmark (lagged)	0.004	0.011	0.005	0.002
Log assets (lagged)	23.357	1.281	23.563	23.017
Bonds outstanding / assets (lagged)	0.075	0.081	0.067	0.089
Total debt / assets (lagged)	0.281	0.137	0.288	0.268
Total liabilities / assets (lagged)	0.660	0.149	0.645	0.685
Cash / assets (lagged)	0.083	0.065	0.075	0.097
Gross profit / assets (lagged)	0.289	0.158	0.289	0.288
Labour cost / assets (lagged)	0.158	0.109	0.145	0.180
SG&A / assets (lagged)	0.182	0.118	0.173	0.198
Issuance dummy	0.559	0.497	0.587	0.513
Eligible-issuance dummy	0.272	0.445	0.311	0.208
Pays dividends	0.968	0.175	0.980	0.949

Bond Sample: Summary Statistics

Back

	Estimation sample		Mean by group	
	Mean	SD	Purchased	Not purch.
Panel A: bond characteristics (4255 bonds, 657 issuers)				
Yield to maturity (%)	2.783	1.888	1.799	3.011
Duration (days)	3000.663	2187.311	2952.225	3010.494
Log duration	7.843	0.561	7.876	7.836
Amount issued (EUR m)	622.785	301.743	743.753	599.040
Log amount issued	20.096	0.631	20.365	20.043
Purchased at issuance (share)	0.164	–	1.000	0.000
Panel B: rating-class distribution				
	N	Share of sample	Purch. within	
AAA	11	0.003	0.000	
AA	182	0.043	0.368	
A	651	0.153	0.270	
BBB	807	0.190	0.321	
BB	248	0.058	0.048	
B	92	0.022	0.000	
CCC and below	12	0.003	0.000	
NR (explicit not-rated grade)	2252	0.529	0.082	

CSPP Holdings: Primary and Secondary Market

Back

	Primary (EUR bn)	Secondary (EUR bn)	Primary share of holdings	Primary share of change in holdings
Jun 2016	0.2	6.2	3.8%	—
Dec 2016	6.9	44.1	13.6%	+15.0%
Dec 2017	19.9	111.7	15.1%	+16.1%
Dec 2018	31.2	146.9	17.5%	+24.2%
Dec 2019	33.5	151.0	18.2%	+35.9%
Dec 2020	52.4	198.0	20.9%	+28.8%
Dec 2021	70.7	239.0	22.8%	+30.8%
Dec 2022	81.3	262.9	23.6%	+30.6%
Dec 2023	75.2	248.8	23.2%	−30.2%
Jul 2024	70.8	232.2	23.4%	−21.0%

Source: ECB, breakdown of CSPP holdings (amortised cost, net of redemptions). Share of increase: change in primary-market holdings over change in total holdings since the previous row.

Eligible Issues and Issuance-Week Purchases, by Year

Back

	Issues	Purchased	Share
2017 (from 1 July)	109	83	0.761
2018	235	162	0.689
2019	299	29	0.097
2020	411	308	0.749
2021	309	204	0.660
2022	277	159	0.574
2023	291	80	0.275
2024	321	58	0.181
Total	2,252	1,083	0.481

Selection into Purchase: All Characteristics

Back

	Across firms	Within firm
	Country, Sector×Year FE	Firm, Year FE
Benchmark share	8.4292***	5.9164
	(3.1409)	(6.6361)
Log assets	0.0652***	-0.0193
	(0.0187)	(0.0490)
Bonds outstanding / assets	0.3971**	-0.6752***
	(0.1976)	(0.2281)
Total debt / assets	0.1642	-0.4921
	(0.1864)	(0.3682)
Total liabilities / assets	-0.4020**	0.0790
	(0.1691)	(0.3094)
Cash / assets	-0.5070*	-0.4331
	(0.2619)	(0.3018)
Gross profit / assets	-0.0372	-0.5242
	(0.2100)	(0.3464)
Labour costs / assets	-0.0178	-0.8999*
	(0.1975)	(0.4643)
SG&A / assets	0.3350	1.0118*
	(0.2623)	(0.5737)
Log markup	0.0766	0.1091
	(0.0615)	(0.1493)
Investment rate	0.4001	0.2892
	(0.6413)	(0.8545)
R^2	0.3615	0.3456
Within R^2	0.0860	0.0197
Observations	1149	1219

Bond Identification

Back

- **A pricing effect:** yield fixed at pricing; no ECB purchase before its bond's issue date $\Rightarrow$ effect on the contracted terms
- **No coupon control:** yield–coupon correlation $= 0.9855 \Rightarrow$ outcome under another name Coupon control
- **Eligibility:** premium large early in sample, near zero during QE $\Rightarrow$ eligible $\times$ year FE (pooled control rejected, $p \leq 0.0001$)
- **"Hot deals" story:** predicts estimate shrinking as cells tighten; here it strengthens, up to same issuer $\times$ same year comparisons
- **Identified residual concerns push the estimate towards zero:**
  - weekly snapshots: bonds allocated at the deal but settled later coded as untreated (about 4.4%)
  - size possibly a mediator: conditioning on it removes part of the effect
  - pre-programme bonds kept in the sample: treatable-bond sample moves estimates away from zero Restricted sample
  - segment-wide effects of the programme: controls partially treated

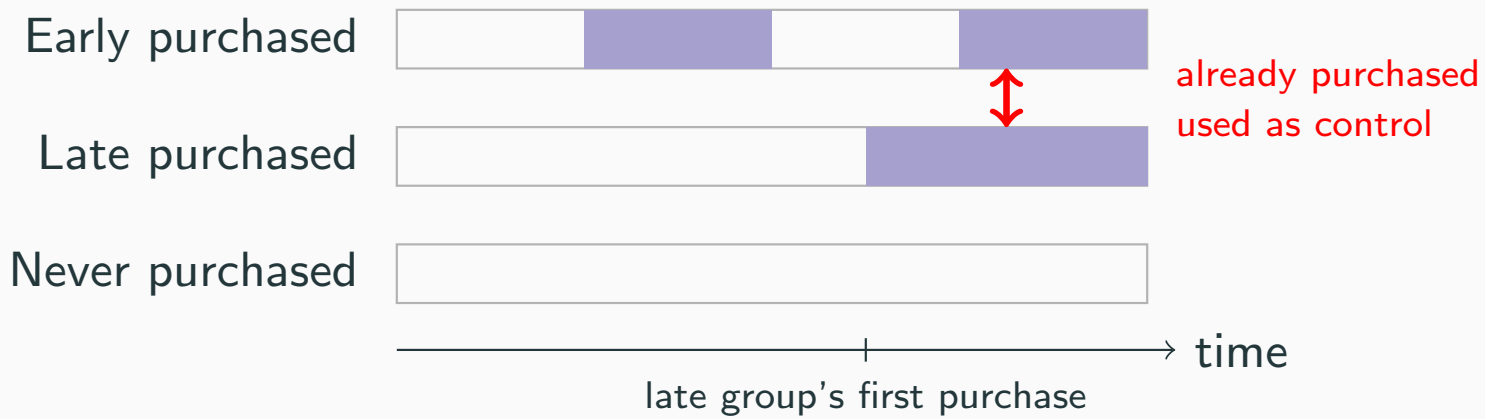
$\Rightarrow$ **Estimates to be read as conservative**

LP-DiD Estimator

Back

Problem:

- Two-way FE fail with staggered, heterogeneous, dynamic effects (e.g. De Chaisemartin and d'Haultfoeuille, 2023; Goodman-Bacon, 2021)
- "Unclean" comparisons: already-treated units used as controls for newly treated ones

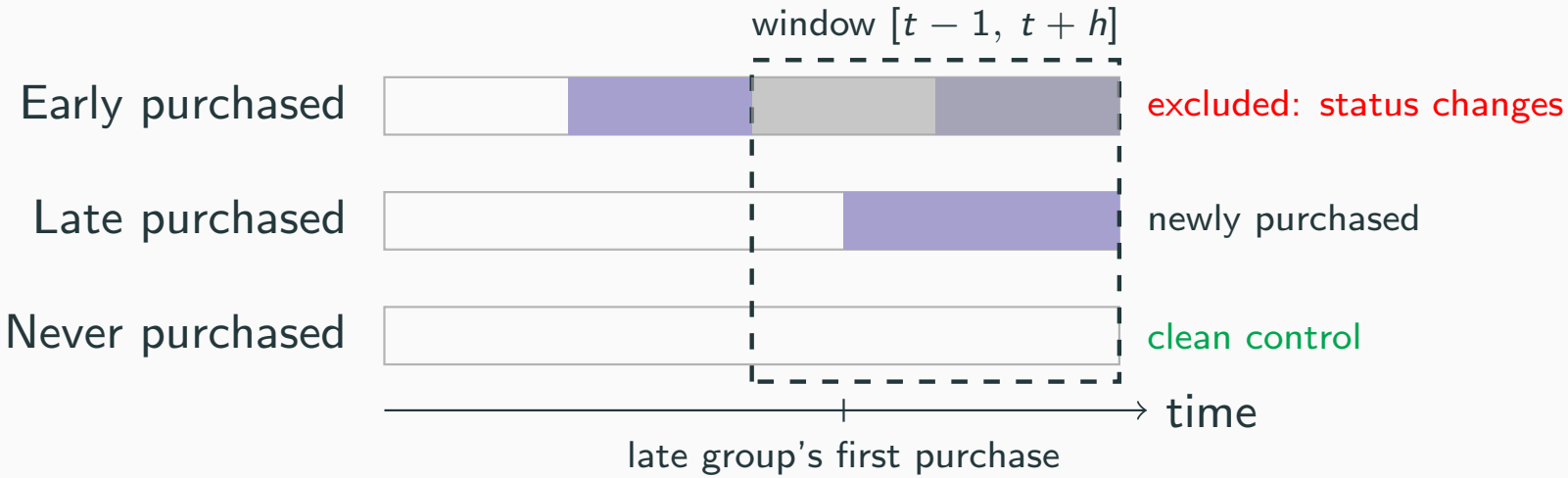


LP-DiD Estimator

Back

Solution: LP-DiD (Dube et al., 2025), non-absorbing version

- Local projections (Jordà, 2005) with clean controls (Cengiz et al., 2019)
- Each horizon: newly purchased vs no change in purchase status over the window



Markup Estimation Strategy

Back

General procedure:

- Data preparation: clean and deflate variables (using 2-digit GVA deflator and country GDP deflator when not available);
- Variables: Gross PPE for capital and COGS for variable input;
- Winsorise all variables at 1% level before.

Estimation procedure in detail:

- First step: $\phi_{i,t} = \sum_{a=0}^{3} \sum_{b=0}^{3} \beta_{ab} \, v_{i,t}^{a} \, k_{i,t}^{b} + \alpha_t + \varepsilon_{i,t}$
- Moment condition: Follow Diez et al. (2021) in adding contemporaneous COGS to deal with ACF critique:

$$\mathbb{E}\left(\xi_{it}(\beta) \begin{pmatrix} \nu_{i,t-1} \\ k_{i,t} \\ \nu_{i,t} \end{pmatrix} \right) = 0$$

Markup Effect: Revenue and Cost Components

Back

Pooled post ($h = 0$–2)	Coefficient	SE	p
Log markup (headline)	0.069^{**}	(0.030)	0.022
ln revenue	0.098	(0.159)	0.539
$-\varepsilon$, De Loecker–Warzynski correction	0.021	(0.045)	0.638
$-$ ln COGS	0.008	(0.161)	0.962
Uncorrected ratio, ln revenue $-$ ln COGS	0.091^{*}	(0.053)	0.088
Observations		592	

- Markup 0.069 = revenue 0.098 − COGS 0.008 − correction 0.021: both components move as the mechanism implies
- Neither component separately significant: common scale shocks cancel only in the ratio

Testing Direct Effects (2)

Back

Yield to Maturity	(1)	(2)	(3)	(4)	(5)
Treated (issuance week)	-0.0769	-0.0117	-0.1694	-0.1976**	-0.2253*
	(0.0870)	(0.0786)	(0.1028)	(0.0826)	(0.1173)
Controls	Yes	Yes	Yes	Yes	Yes
Issuer FE	✓	✓		✓	✓
Issuer × Year FE			✓		
NACE × Year FE				✓	
Country × Year FE				✓	
NACE × Country × Year FE					✓
Eligible × Year FE	✓	✓	✓	✓	✓
Year FE	✓				
Quarter FE		✓			
Bond Rating FE	✓	✓	✓	✓	✓
Bond Type FE				✓	✓
Coupon Type FE				✓	✓
Observations	3,293	3,293	2,217	2,918	2,402
R^2	0.8541	0.8798	0.9379	0.9101	0.9296
Adjusted R^2	0.8299	0.8576	0.8989	0.8599	0.8737

Yields — Coupon-Rate Control

Yield to Maturity	(1)	(2)	(3)	(4)
Treated (issuance week)	-0.0239	-0.0154	-0.0245	-0.0334
	(0.0239)	(0.0232)	(0.0292)	(0.0311)
Controls	Yes	Yes	Yes	Yes
Coupon	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓
NACE × Year FE	✓	✓		
Country × Year FE	✓	✓		
NACE × Country × Year FE			✓	✓
Eligible × Year FE	✓	✓	✓	✓
Bond Rating FE	✓		✓	
Rating × Year FE		✓		✓
Observations	2,909	2,864	2,400	2,356
R^2	0.9916	0.9922	0.9911	0.9917
Adjusted R^2	0.9869	0.9870	0.9842	0.9839

Yields — Robustness Battery

Check (column (1) specification)	Coefficient	Standard error	Bonds
Baseline: headline	-0.2170***	(0.0809)	2,920
Prior-purchase controls, issuance-week purchases, one-year window	-0.2152***	(0.0815)	2,920
Prior-purchase controls, issuance-week purchases, one- and two-year windows	-0.2175***	(0.0807)	2,920
Prior-purchase controls, purchases of any kind, one-year window	-0.2169***	(0.0813)	2,920
Prior-purchase controls, purchases of any kind, one- and two-year windows	-0.2221***	(0.0813)	2,920
Clustering by issuer and year-quarter	-0.2170**	(0.0936)	2,920
Clustering by issuer and year	-0.2170**	(0.0771)	2,920
Rating classes	-0.2060**	(0.0826)	2,920
Fixed-coupon bonds	-0.1920**	(0.0838)	2,841
Post-programme sample	-0.2658***	(0.0834)	2,000
Alternative yield measures, recomputed yield	-0.2210***	(0.0751)	3,022
Alternative yield measures, unified yield	-0.2435***	(0.0724)	3,098
Benchmark without commercial paper	-0.2171***	(0.0808)	2,920
Treatment including the initial portfolio	-0.2618***	(0.0788)	2,920

Testing Issuing Conditions: Robustness Checks

Back

- Previous purchases: conditioning on previous issuance-week purchases, and on previous purchases of any kind incl. benchmark bonds (1-year and 1–2-year windows) Primary 1y Primary 1–2y Any 1y Any 1–2y
- Clustering: two-way, on issuer and year-quarter, and on issuer and year Year-quarter Year
- Inference: wild cluster bootstrap and randomisation inference Bootstrap Randomisation
- Rating: broad rating dummies instead of the rating fixed effect Results
- Sample: fixed-coupon bonds only; bonds issued after the programme start Fixed coupon Post-programme
- Measurement: two alternative yield measures; CSPP benchmark excluding commercial paper Yield 1 Yield 2 Benchmark
- Treatment: including benchmark-portfolio bonds Results
- Eligibility: four treatments of CSPP eligibility Results
- Coupon rate: controlling for it, a bad control Results

Yields — Previous Primary Purchases (1y)

Back

Yield to Maturity	(1)	(2)	(3)	(4)
Treated (issuance week)	-0.2152***	-0.1900**	-0.2520**	-0.2368*
	(0.0815)	(0.0903)	(0.1159)	(0.1346)
Prev. primary purchase (1y)	0.0312	0.0592	-0.0256	0.0318
	(0.0911)	(0.0967)	(0.1328)	(0.1485)
Treatment	Issuance week, observed			
Controls	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓
NACE × Year FE	✓	✓		
Country × Year FE	✓	✓		
NACE × Country × Year FE			✓	✓
Eligible × Year FE	✓	✓	✓	✓
Bond Rating FE	✓		✓	
Rating × Year FE		✓		✓
Observations	2,920	2,873	2,405	2,361
R^2	0.9075	0.9147	0.9267	0.9336
Adjusted R^2	0.8563	0.8587	0.8690	0.8715

Yields — Previous Primary Purchases (1–2y)

Back

Yield to Maturity	(1)	(2)	(3)	(4)
Treated (issuance week)	-0.2175***	-0.1920**	-0.2616**	-0.2403*
	(0.0807)	(0.0899)	(0.1127)	(0.1328)
Prev. primary purchase (1y)	0.0323	0.0597	-0.0309	0.0281
	(0.0907)	(0.0963)	(0.1292)	(0.1430)
Prev. primary purchase (2y)	-0.0793	-0.0569	-0.0920	-0.0415
	(0.0983)	(0.0974)	(0.1464)	(0.1608)
Treatment	Issuance week, observed			
Controls	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓
NACE × Year FE	✓	✓		
Country × Year FE	✓	✓		
NACE × Country × Year FE			✓	✓
Eligible × Year FE	✓	✓	✓	✓
Bond Rating FE	✓		✓	
Rating × Year FE		✓		✓
Observations	2,920	2,873	2,405	2,361
R^2	0.9076	0.9147	0.9268	0.9336
Adjusted R^2	0.8564	0.8586	0.8690	0.8715

Yields — Previous Purchases, Any Kind (1y)

Back

Yield to Maturity	(1)	(2)	(3)	(4)
Treated (issuance week)	-0.2169***	-0.1938**	-0.2485**	-0.2432*
	(0.0813)	(0.0909)	(0.1144)	(0.1327)
Prev. purchase, any (1y)	0.0010	0.0181	-0.0686	-0.0141
	(0.0795)	(0.0811)	(0.1188)	(0.1302)
Treatment	Issuance week, observed			
Controls	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓
NACE × Year FE	✓	✓		
Country × Year FE	✓	✓		
NACE × Country × Year FE			✓	✓
Eligible × Year FE	✓	✓	✓	✓
Bond Rating FE	✓		✓	
Rating × Year FE		✓		✓
Observations	2,920	2,873	2,405	2,361
R^2	0.9075	0.9146	0.9267	0.9336
Adjusted R^2	0.8563	0.8586	0.8690	0.8715

Yields — Previous Purchases, Any Kind (1–2y)

Back

Yield to Maturity	(1)	(2)	(3)	(4)
Treated (issuance week)	-0.2221***	-0.2003**	-0.2518**	-0.2508*
	(0.0813)	(0.0900)	(0.1133)	(0.1290)
Prev. purchase, any (1y)	-0.0010	0.0155	-0.0669	-0.0096
	(0.0795)	(0.0816)	(0.1191)	(0.1307)
Prev. purchase, any (2y)	0.0510	0.0623	0.0585	0.0843
	(0.0590)	(0.0611)	(0.0919)	(0.1070)
Treatment	Issuance week, observed			
Controls	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓
NACE × Year FE	✓	✓		
Country × Year FE	✓	✓		
NACE × Country × Year FE			✓	✓
Eligible × Year FE	✓	✓	✓	✓
Bond Rating FE	✓		✓	
Rating × Year FE		✓		✓
Observations	2,920	2,873	2,405	2,361
R^2	0.9075	0.9147	0.9268	0.9337
Adjusted R^2	0.8563	0.8586	0.8690	0.8716

Yields — Clustering: Issuer, Year-Quarter

Back

Yield to Maturity	(1)	(2)	(3)	(4)
Treated (issuance week)	-0.2170**	-0.1948*	-0.2484**	-0.2427*
	(0.0936)	(0.1036)	(0.1180)	(0.1323)
Treatment	Issuance week, observed			
Controls	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓
NACE × Year FE	✓	✓		
Country × Year FE	✓	✓		
NACE × Country × Year FE			✓	✓
Eligible × Year FE	✓	✓	✓	✓
Bond Rating FE	✓		✓	
Rating × Year FE		✓		✓
Observations	2,920	2,873	2,405	2,361
R^2	0.9075	0.9146	0.9267	0.9336
Adjusted R^2	0.8564	0.8587	0.8690	0.8716

Yields — Clustering: Issuer, Year

Back

Yield to Maturity	(1)	(2)	(3)	(4)
Treated (issuance week)	-0.2170**	-0.1948**	-0.2484***	-0.2427**
	(0.0771)	(0.0882)	(0.0774)	(0.0867)
Treatment	Issuance week, observed			
Controls	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓
NACE × Year FE	✓	✓		
Country × Year FE	✓	✓		
NACE × Country × Year FE			✓	✓
Eligible × Year FE	✓	✓	✓	✓
Bond Rating FE	✓		✓	
Rating × Year FE		✓		✓
Observations	2,920	2,873	2,405	2,361
R^2	0.9075	0.9146	0.9267	0.9336
Adjusted R^2	0.8537	0.8544	0.8674	0.8673

Yields — Wild Cluster Bootstrap

Back

	Analytical (cluster-robust, t_{G-1})	Wild cluster bootstrap (Rademacher, 9,999 reps)
Coefficient on purchase	-0.2170	-0.2170
Standard error	0.0809	—
t-statistic	-2.680	-2.680
p-value	0.0077	0.0106
95% confidence interval	[-0.3761, -0.0578]	[-0.3770, -0.0587]
90% confidence interval	[-0.3504, -0.0835]	[-0.3517, -0.0842]
Observations	2,920	2,920
Clusters (issuer groups)	354	354

Yields — Randomisation Inference

Back

Panel A. Observed estimate	
Coefficient on purchase	-0.2170
Cluster-robust standard error	0.0809
t-statistic	-2.680
Analytical p-value	0.0077
Panel B. Randomization inference	
Replications	5,000
Permutations with $\lvert b \rvert$ at least the observed	0
RI p-value, two-sided	0.0000
small-sample form $(1 + n)/(1 + R)$	0.0002
RI p-value, one-sided	0.0000
RI p-value on the t-statistic, two-sided	0.0070

Yields — Broad Rating Dummies

Back

Yield to Maturity	(1)	(2)
Treated (issuance week)	-0.2060**	-0.2411**
	(0.0826)	(0.1166)
Treatment	Issuance week, observed	
Controls	Yes	Yes
Issuer FE	✓	✓
NACE × Year FE	✓	
Country × Year FE	✓	
NACE × Country × Year FE		✓
Eligible × Year FE	✓	✓
Observations	2,920	2,407
R^2	0.9070	0.9257
Adjusted R^2	0.8564	0.8682

Yields — Fixed-Coupon Bonds Only

Back

Yield to Maturity	(1)	(2)	(3)	(4)
Treated (issuance week)	-0.1920**	-0.1773*	-0.2236*	-0.2199
	(0.0838)	(0.0928)	(0.1205)	(0.1417)
Treatment	Issuance week, observed			
Controls	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓
NACE × Year FE	✓	✓		
Country × Year FE	✓	✓		
NACE × Country × Year FE			✓	✓
Eligible × Year FE	✓	✓	✓	✓
Bond Rating FE	✓		✓	
Rating × Year FE		✓		✓
Sample	Fixed-coupon	Fixed-coupon	Fixed-coupon	Fixed-coupon
Observations	2,841	2,791	2,327	2,274
R^2	0.9065	0.9128	0.9242	0.9290
Adjusted R^2	0.8539	0.8546	0.8641	0.8627

Yields — Bonds Issued After Programme Start

Back

Yield to Maturity	(1)	(2)	(3)	(4)
Treated (issuance week)	-0.2658***	-0.2446***	-0.2580**	-0.2632**
	(0.0834)	(0.0942)	(0.1128)	(0.1247)
Treatment	Issuance week, observed			
Controls	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓
NACE × Year FE	✓	✓		
Country × Year FE	✓	✓		
NACE × Country × Year FE			✓	✓
Eligible × Year FE	✓	✓	✓	✓
Bond Rating FE	✓		✓	
Rating × Year FE		✓		✓
Sample	Post-Jun 2016	Post-Jun 2016	Post-Jun 2016	Post-Jun 2016
Observations	2,000	1,983	1,684	1,662
R^2	0.9254	0.9365	0.9388	0.9471
Adjusted R^2	0.8799	0.8896	0.8860	0.8926

Yields — Yield Measure: ytm_calc

Back

Yield to Maturity (recomputed)	(1)	(2)	(3)	(4)
Treated (issuance week)	-0.2210***	-0.2072**	-0.2673***	-0.2633**
	(0.0751)	(0.0824)	(0.0990)	(0.1146)
Treatment	Issuance week, observed			
Controls	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓
NACE × Year FE	✓	✓		
Country × Year FE	✓	✓		
NACE × Country × Year FE			✓	✓
Eligible × Year FE	✓	✓	✓	✓
Bond Rating FE	✓		✓	
Rating × Year FE		✓		✓
Observations	3,022	2,977	2,467	2,422
R^2	0.9016	0.9097	0.9190	0.9247
Adjusted R^2	0.8472	0.8506	0.8556	0.8550

Yields — Yield Measure: ytm_unified

Back

Yield to Maturity (unified)	(1)	(2)	(3)	(4)
Treated (issuance week)	-0.2435***	-0.2223***	-0.2940***	-0.2732**
	(0.0724)	(0.0791)	(0.0947)	(0.1060)
Treatment	Issuance week, observed			
Controls	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓
NACE × Year FE	✓	✓		
Country × Year FE	✓	✓		
NACE × Country × Year FE			✓	✓
Eligible × Year FE	✓	✓	✓	✓
Bond Rating FE	✓		✓	
Rating × Year FE		✓		✓
Observations	3,098	3,055	2,536	2,494
R^2	0.9015	0.9100	0.9202	0.9270
Adjusted R^2	0.8477	0.8521	0.8582	0.8599

Yields — Benchmark Excluding CP

Back

Yield to Maturity	(1)	(2)	(3)	(4)
Treated (issuance week)	-0.2171***	-0.1950**	-0.2492**	-0.2435*
	(0.0808)	(0.0900)	(0.1158)	(0.1323)
Treatment	Issuance week, observed			
Controls	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓
NACE × Year FE	✓	✓		
Country × Year FE	✓	✓		
NACE × Country × Year FE			✓	✓
Eligible × Year FE	✓	✓	✓	✓
Bond Rating FE	✓		✓	
Rating × Year FE		✓		✓
Observations	2,920	2,873	2,405	2,361
R^2	0.9075	0.9146	0.9267	0.9336
Adjusted R^2	0.8563	0.8586	0.8690	0.8716

Yields — Treatment Incl. Benchmark Bonds

Back

Yield to Maturity	(1)	(2)	(3)	(4)
Treated (incl. benchmark)	-0.2618***	-0.2300***	-0.2785**	-0.2785**
	(0.0788)	(0.0877)	(0.1095)	(0.1203)
Treatment	Issuance week, incl. benchmark (imputed)			
Controls	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓
NACE × Year FE	✓	✓		
Country × Year FE	✓	✓		
NACE × Country × Year FE			✓	✓
Eligible × Year FE	✓	✓	✓	✓
Bond Rating FE	✓		✓	
Rating × Year FE		✓		✓
Observations	2,920	2,873	2,405	2,361
R^2	0.9078	0.9148	0.9269	0.9338
Adjusted R^2	0.8568	0.8590	0.8694	0.8720

Yields — Four Treatments of Eligibility

Yield to Maturity	(1)	(2)	(3)	(4)
Treated (issuance week)	-0.0720	-0.1333	-0.1274	-0.2170***
	(0.0952)	(0.0826)	(0.0840)	(0.0809)
eligible	-0.4511**		-0.6687**	
	(0.1867)		(0.2760)	
Eligible and issued post-announcement		-0.1946	0.3551*	
		(0.1241)	(0.2034)	
Controls	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓
NACE × Year FE	✓	✓	✓	✓
Country × Year FE	✓	✓	✓	✓
Eligible × Year FE				✓
Bond Rating FE	✓	✓	✓	✓
Observations	2,920	2,920	2,920	2,920
R^2	0.9044	0.9024	0.9049	0.9075
Adjusted R^2	0.8526	0.8496	0.8533	0.8564

Yields — Spillovers: Full Sample

Yield to Maturity	(1)	(2)	(3)	(4)
Prev. purchase, any (1y)	-0.0798	0.0010	0.0181	-0.0686
	(0.0617)	(0.0795)	(0.0811)	(0.1188)
Treated (issuance week)	-0.0818	-0.2169***	-0.1938**	-0.2485**
	(0.0863)	(0.0813)	(0.0909)	(0.1144)
Controls	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓
NACE × Year FE		✓	✓	
Country × Year FE		✓	✓	
NACE × Country × Year FE				✓
Eligible × Year FE	✓	✓	✓	✓
Year FE	✓			
Bond Rating FE	✓	✓		✓
Rating × Year FE			✓	
Sample	Full	Full	Full	Full
Observations	3,293	2,920	2,873	2,405
R^2	0.8542	0.9075	0.9146	0.9267
Adjusted R^2	0.8300	0.8563	0.8586	0.8690

Yields — Spillovers: Never-Purchased Sample

Back

Yield to Maturity	(1)	(2)	(3)	(4)
Prev. purchase, any (1y)	-0.1686**	-0.1289	-0.0655	-0.2765**
	(0.0813)	(0.0983)	(0.1046)	(0.1349)
Controls	Yes	Yes	Yes	Yes
Issuer FE	✓	✓	✓	✓
NACE × Year FE		✓	✓	
Country × Year FE		✓	✓	
NACE × Country × Year FE				✓
Eligible × Year FE	✓	✓	✓	✓
Year FE	✓			
Bond Rating FE	✓	✓		✓
Rating × Year FE			✓	
Sample	Never-treated	Never-treated	Never-treated	Never-treated
Observations	2,622	2,244	2,189	1,813
R^2	0.8600	0.9216	0.9311	0.9365
Adjusted R^2	0.8316	0.8656	0.8707	0.8763

Capital — Robustness Battery

Perturbation	Pooled post	Cumul. ($\approx \Delta K/K$)	Pooled pre	N
Baseline: headline	2.2**	6.4**	0.0	599
	(0.9)	(2.8)	(0.3)	
FE: sector, country, year	1.5**	4.4**	0.2	727
	(0.7)	(2.2)	(0.2)	
FE: sector + country×year	1.9**	5.7**	0.3	682
	(0.8)	(2.3)	(0.3)	
FE: sector×year + country×year	1.8***	5.5***	-0.1	530
	(0.6)	(1.9)	(0.4)	
Controls: drop issuance dummies	2.2**	6.8**	-0.1	599
	(0.9)	(2.8)	(0.2)	
Control: + lagged treatment	2.2**	6.5**	0.0	599
	(0.9)	(2.7)	(0.3)	
Control: + lagged purchase count	2.2**	6.5**	0.0	599
	(0.9)	(2.7)	(0.3)	
Control: + lagged cumul. purchases	2.2**	6.4**	0.1	599
	(0.9)	(2.8)	(0.3)	
Treatment: benchmark-inclusive purchase	0.5	1.3	-0.0	526
	(0.7)	(2.0)	(0.3)	
COVID: 2020–21 outcome blanked	0.4	–	0.2	799
	(0.4)		(0.2)	
Gate: eligibility excl. commercial paper	2.2**	6.6**	0.0	556
	(1.0)	(2.9)	(0.3)	
Gate: pre-determined eligibility	0.2	0.6	0.1	191
	(1.3)	(3.9)	(0.4)	
AR: lagged outcome level	2.2**	6.4**	-0.2	599
	(0.9)	(2.8)	(0.2)	
AR: lagged outcome first difference	1.5	4.5	-0.1	461
	(1.0)	(2.8)	(0.1)	
$L = 2$	2.2**	6.6**	0.1	586
	(1.0)	(3.1)	(0.3)	
$L = 3$	2.2**	6.5**	0.1	578
	(1.0)	(3.1)	(0.3)	
Variant: first purchase only treated	2.1**	6.3**	0.1	575
	(1.1)	(3.2)	(0.3)	
Variant: not-yet-treated controls	2.2**	6.4**	0.1	576
	(0.9)	(2.8)	(0.3)	
Variant: first purchase + not-yet-treated	1.7*	5.1*	0.1	575
	(1.0)	(2.9)	(0.3)	
Variant: standard LP	0.0	0.0	0.0	1188
	(0.1)	(0.3)	(0.1)	

Testing Capital Channel: Robustness Checks

Back

- Different FEs: year + sector + country, year + sector× year + country × year Results

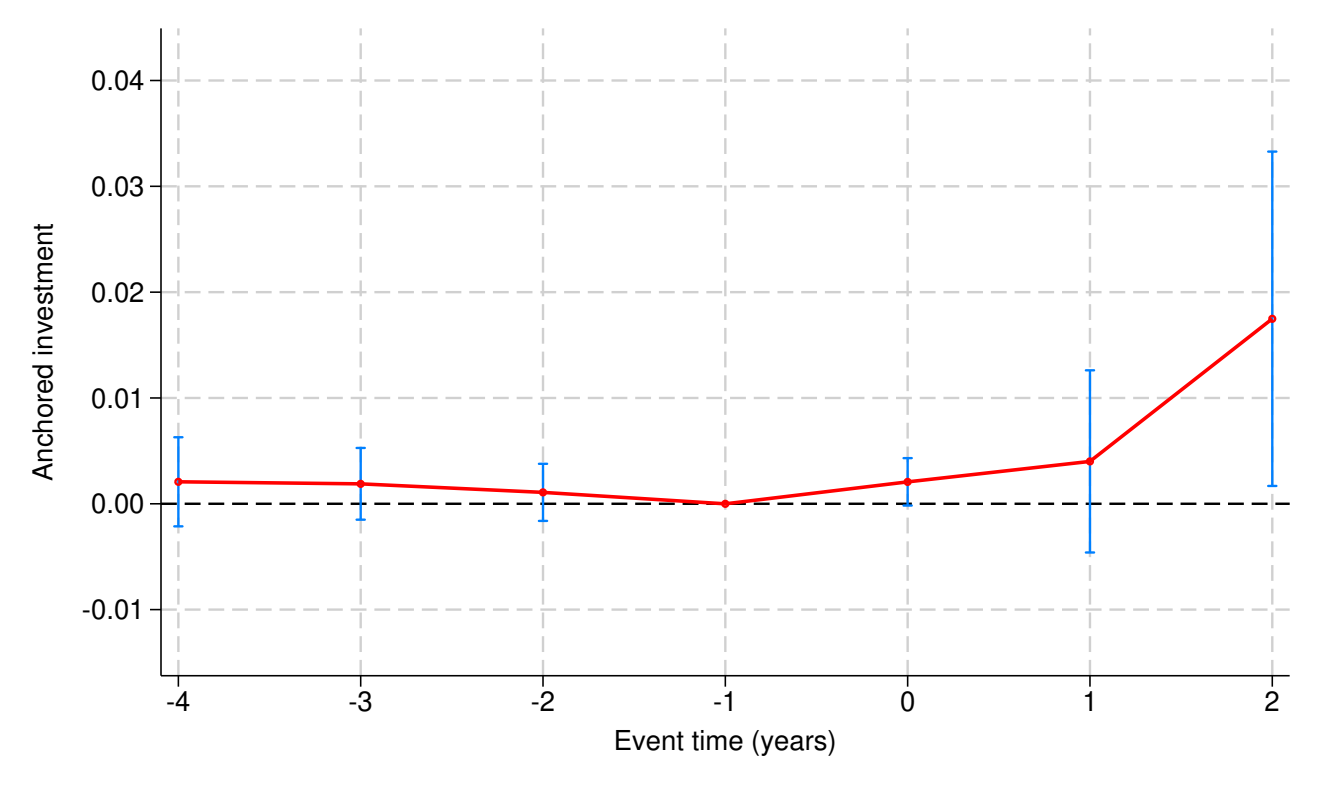
- Different Non-Absorbing options: restricting control group to not-yet treated only, restricting treated group to first time entries only, both combined. Results

- Different value of stabilisation effect parameter L: $L = 2$ and $L = 3$. Results

- Control for autoregressive patterns: control for 1 lag of y variable. Results

Capital — Robustness: Different FEs (sector + country + year)

Back

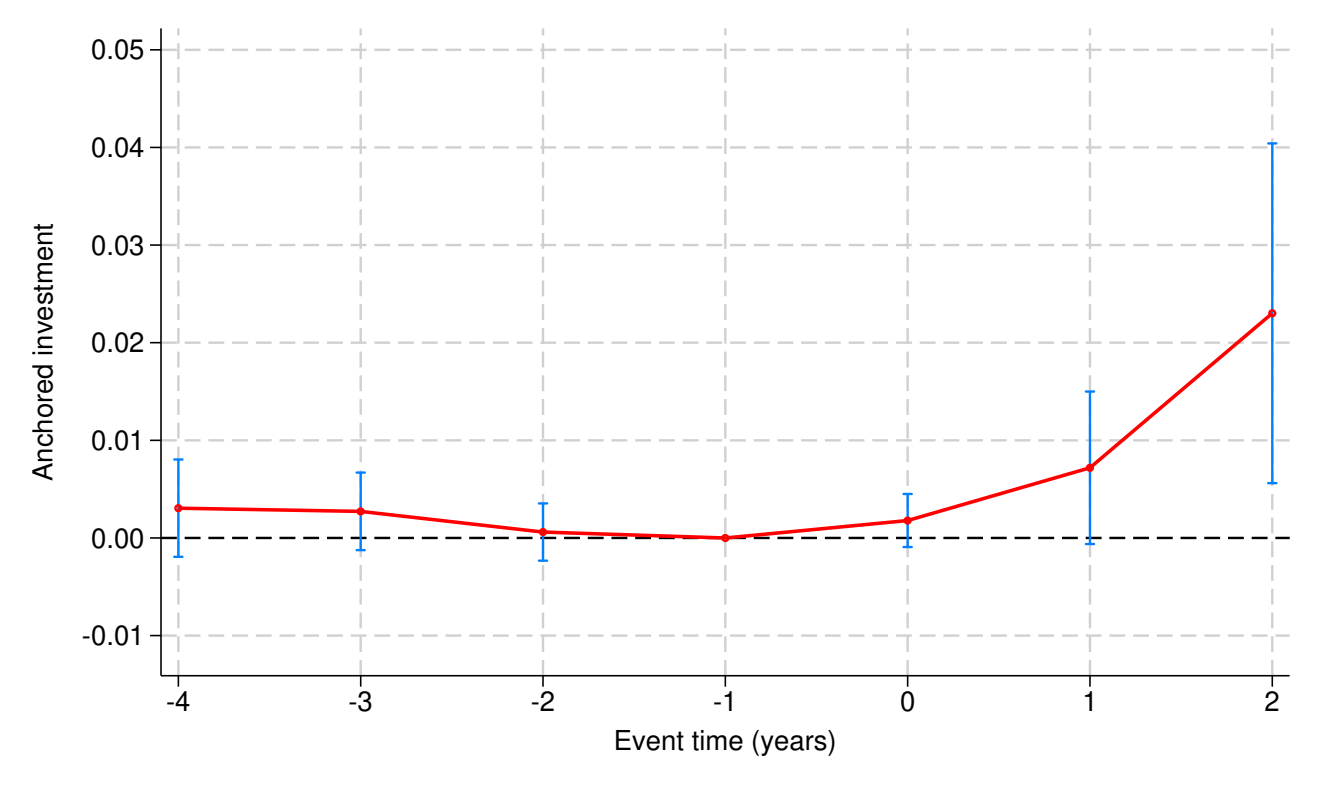


Pooled Estimates

	Estimate	SE	p-value	90% CI		Obs
Pre	0.0019534	0.0020773	0.3487	-0.0014873	0.0053941	646
Post	**0.0153758**	0.0074025	0.0396	0.0031199	0.0276317	727

Capital — Robustness: Different FEs (sector + country × year)

Back

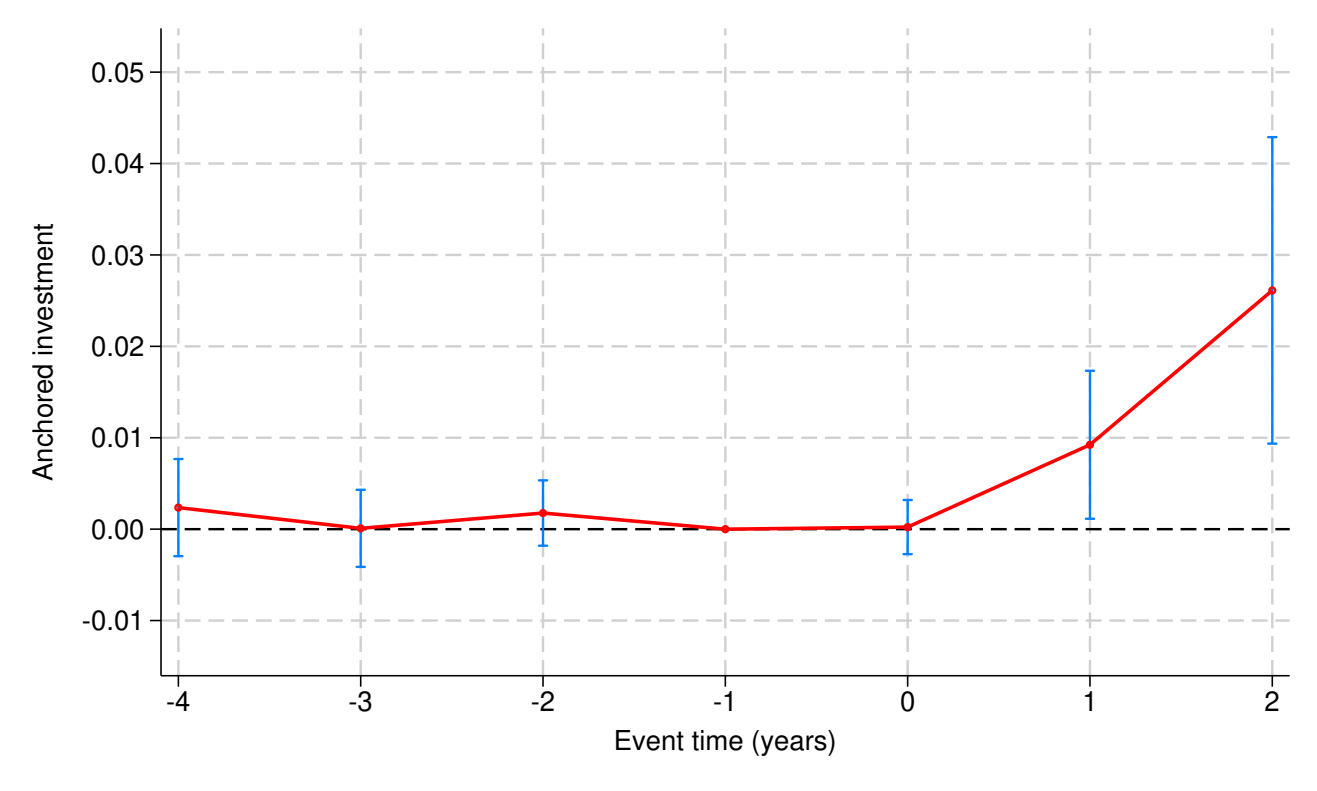


Pooled Estimates

	Estimate	SE	p-value	90% CI		Obs
Pre	0.0026919	0.0026258	0.3072	-0.0016580	0.0070417	611
Post	**0.0188296**	0.0075494	0.0138	0.0063281	0.0313311	682

Capital — Robustness: Non-Abs (controls = not-yet-treated)

Back

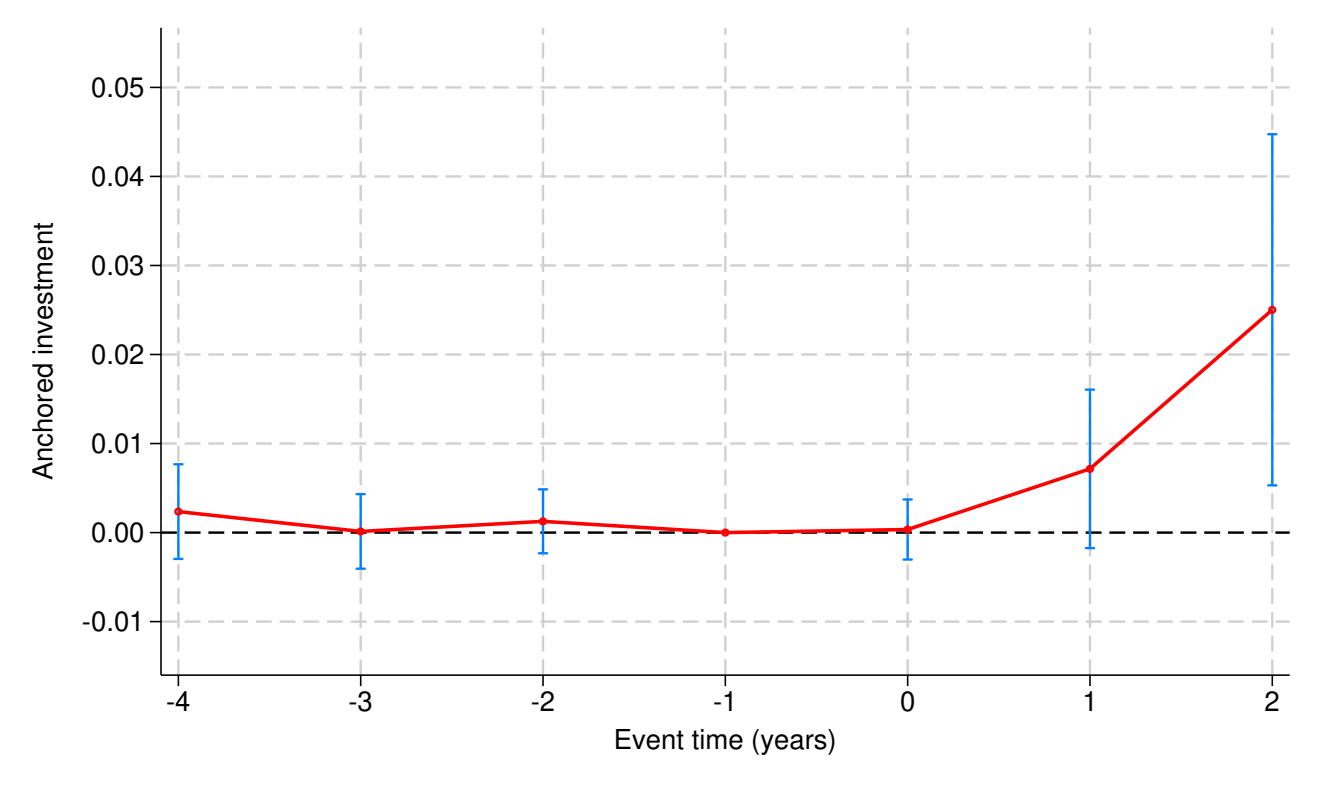


Pooled Estimates

	Estimate	SE	p-value	90% CI		Obs
Pre	0.0006458	0.0028747	0.8226	-0.0041189	0.0054105	527
Post	**0.0215389**	0.0093294	0.0225	0.0060842	0.0369937	576

Capital — Robustness: Non-Abs (treated = first-time)

Back

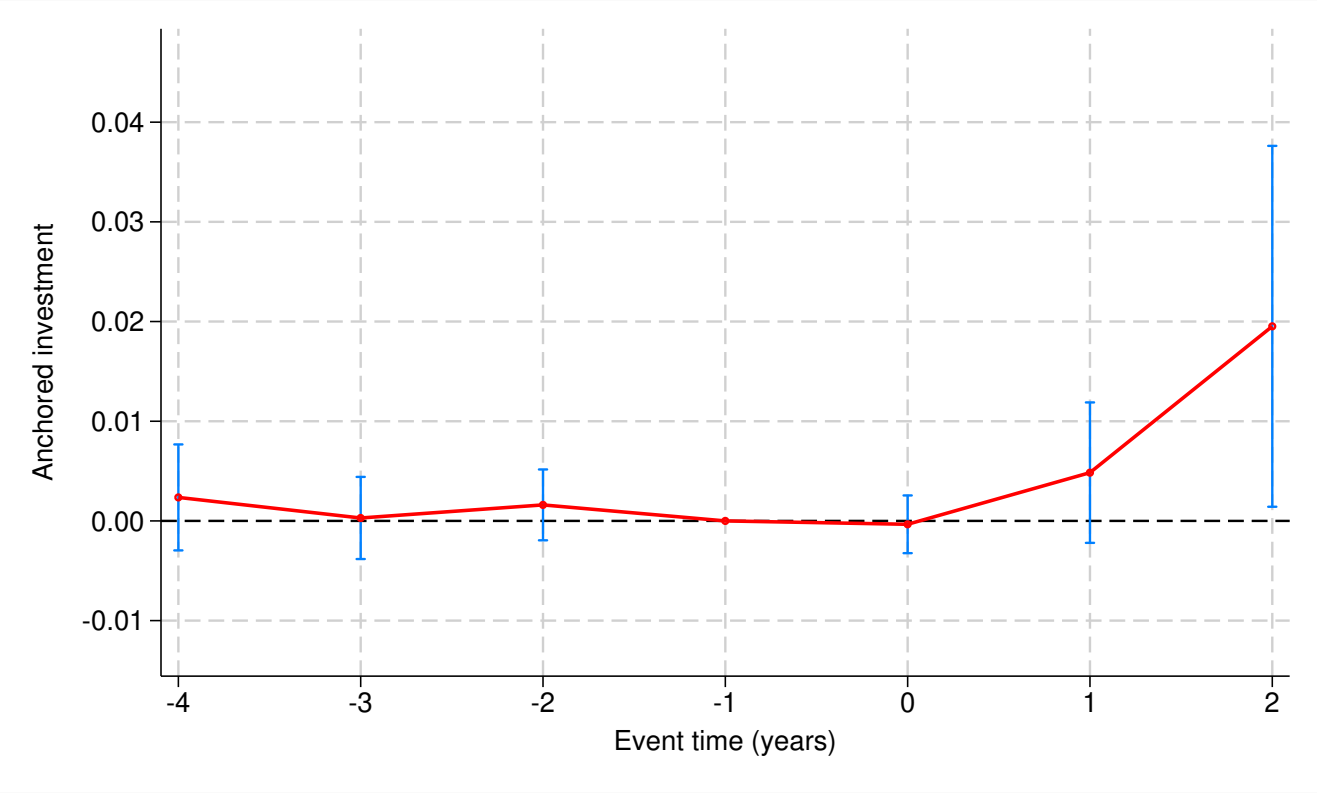


Pooled Estimates

	Estimate	SE	p-value	90% CI		Obs
Pre	0.0006458	0.0028747	0.8226	-0.0041189	0.0054105	527
Post	**0.0214421**	0.0106638	0.0464	0.0037759	0.0391083	575

Capital — Robustness: Non-Abs (both combined)

Back

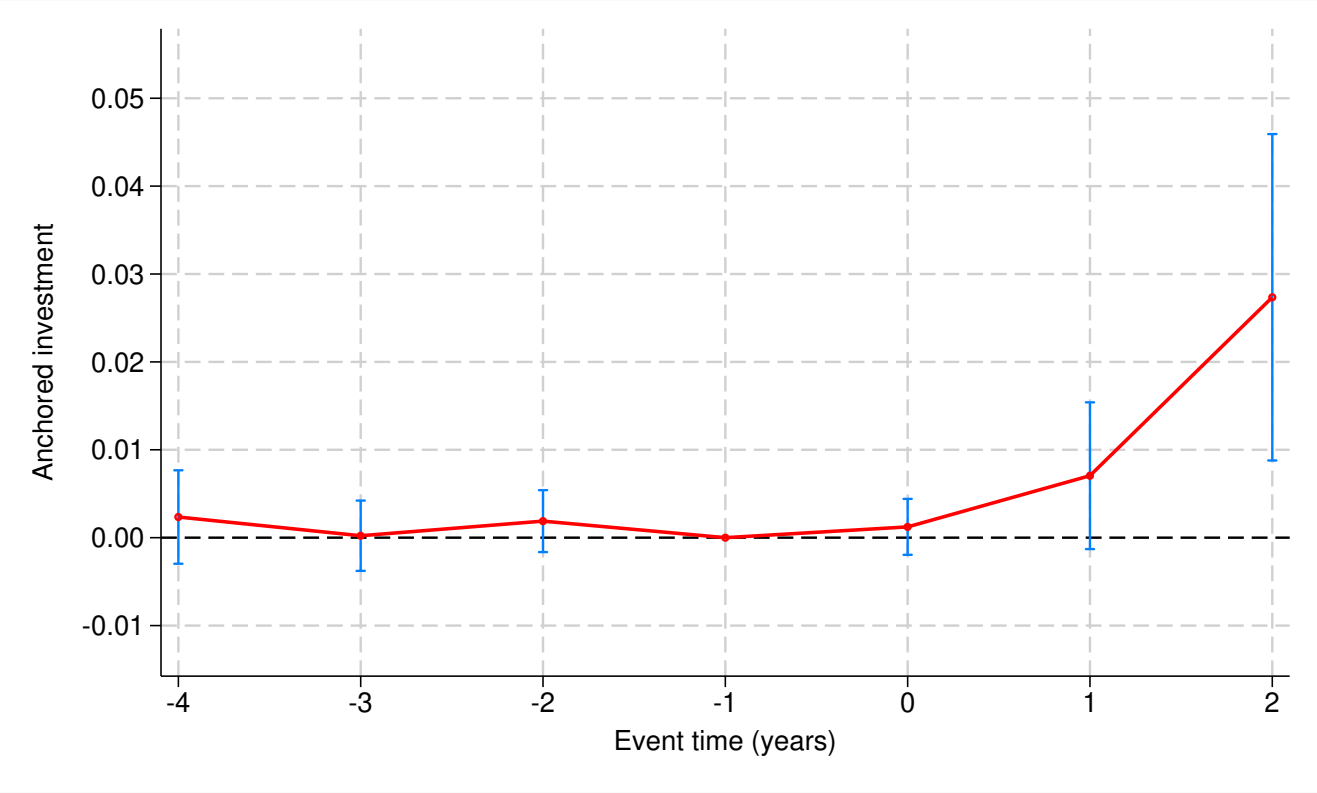


Pooled Estimates

	Estimate	SE	p-value	90% CI		Obs
Pre	0.0006458	0.0028747	0.8226	-0.0041189	0.0054105	527
Post	**0.0172201**	0.0098203	0.0819	0.0009521	0.0334881	575

Capital — Robustness: Stabilisation parameter $L = 2$

Back

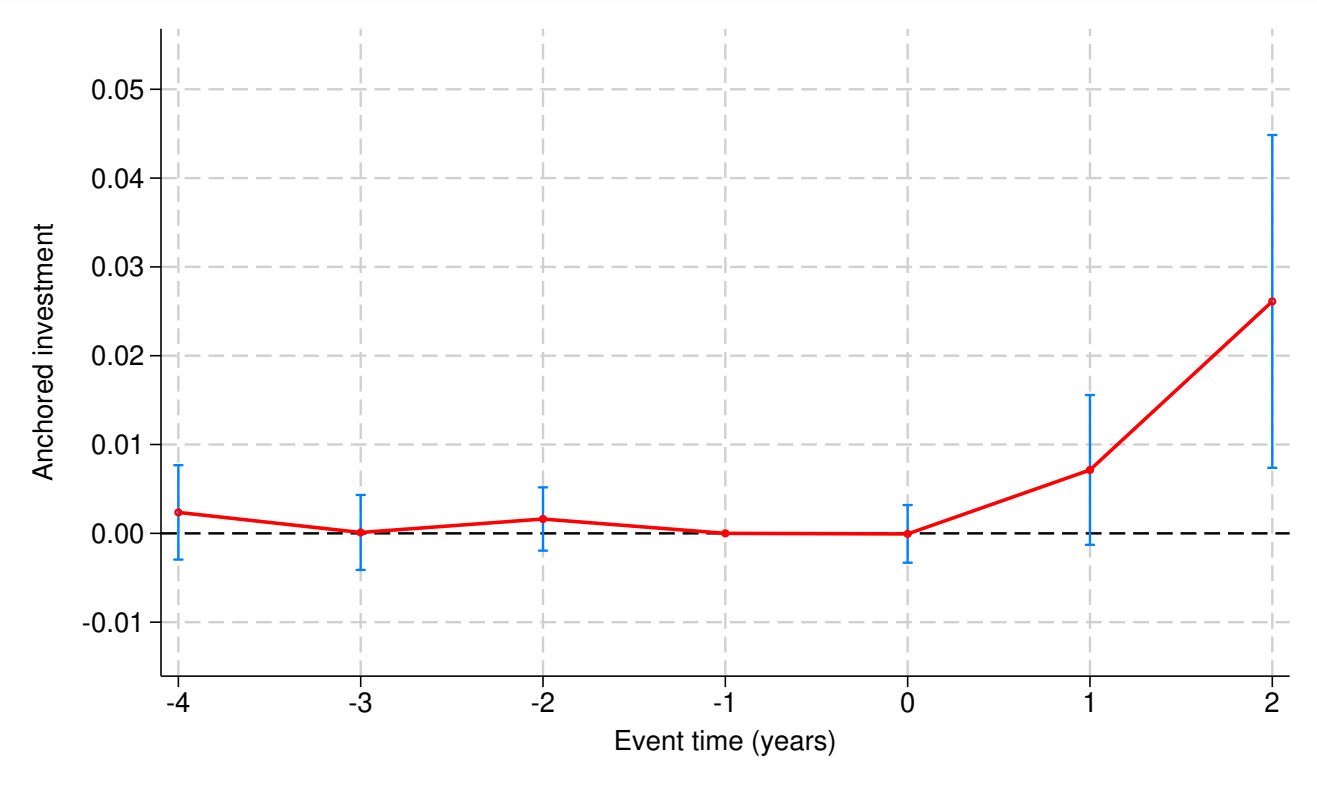


Pooled Estimates

	Estimate	SE	p-value	90% CI		Obs
Pre	0.0006424	0.0028762	0.8236	-0.0041247	0.0054095	529
Post	**0.0224457**	0.0103083	0.0312	0.0053702	0.0395212	586

Capital — Robustness: Stabilisation parameter $L = 3$

Back

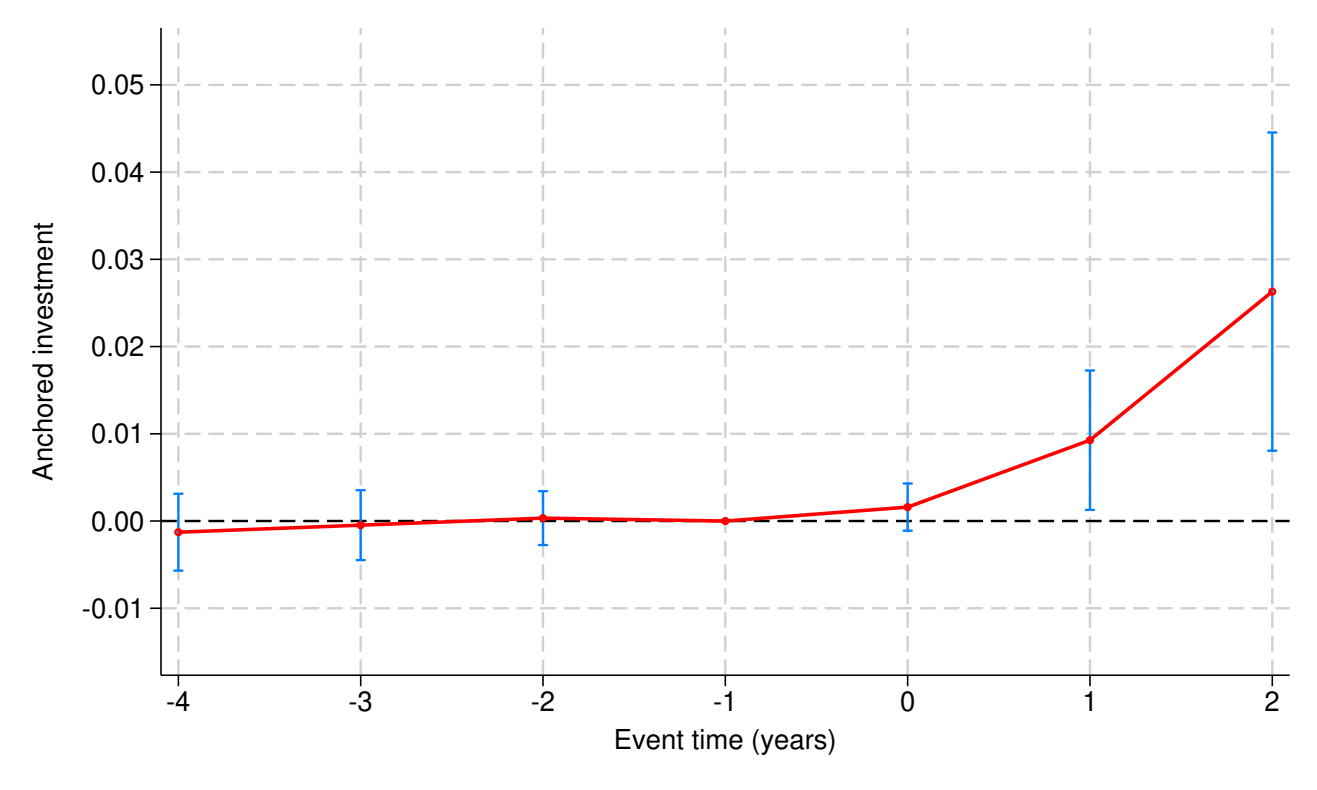


Pooled Estimates

	Estimate	SE	p-value	90% CI		Obs
Pre	0.0006458	0.0028747	0.8226	-0.0041189	0.0054105	527
Post	**0.0220665**	0.0104037	0.0358	0.0048312	0.0393019	578

Capital — Robustness: AR control (1 lag of y)

Back



Pooled Estimates

	Estimate	SE	p-value	90% CI		Obs
Pre	-0.0016912	0.0024262	0.4871	-0.0057121	0.0023296	536
Post	**0.0215633**	0.0094116	0.0235	0.0059741	0.0371525	599

Markups — Robustness Battery

Perturbation	Pooled post	Pooled pre	N
Baseline: headline	6.9**	1.3	592
	(3.0)	(1.4)	
Outcome: log input share	6.9**	1.3	592
	(3.0)	(1.4)	
FE: sector, country, year	6.5**	1.2	716
	(2.6)	(1.6)	
FE: sector + country×year	10.0**	0.5	669
	(3.9)	(1.9)	
FE: sector×year + country×year	10.0	0.2	517
	(6.4)	(2.0)	
Controls: drop issuance dummies	6.7**	1.1	592
	(2.6)	(1.2)	
Control: + lagged treatment	6.9**	1.3	592
	(3.0)	(1.4)	
Control: + lagged purchase count	6.9**	1.3	592
	(3.0)	(1.4)	
Control: + lagged cumul. purchases	6.9**	1.3	592
	(3.0)	(1.4)	
Treatment: benchmark-inclusive purchase	6.0**	0.4	502
	(2.7)	(1.4)	
COVID: 2020–21 outcome blanked	6.8***	3.2	690
	(2.6)	(2.2)	
Gate: eligibility excl. commercial paper	7.1**	1.8	557
	(3.3)	(1.5)	
Gate: pre-determined eligibility	12.4**	3.6	194
	(5.8)	(2.8)	
AR: lagged outcome level	8.2***	1.5	592
	(3.0)	(1.3)	
AR: lagged outcome first difference	9.0***	1.3	454
	(2.8)	(0.9)	
$L = 2$	6.9**	1.4	603
	(2.9)	(1.4)	
$L = 1$	5.1*	1.5	617
	(2.9)	(1.4)	
Variant: first purchase only treated	6.9**	1.3	588
	(3.0)	(1.4)	
Variant: not-yet-treated controls	6.9**	1.3	588
	(3.0)	(1.4)	
Variant: first purchase + not-yet-treated	6.2**	1.3	589
	(2.8)	(1.4)	
Variant: standard LP	1.0*	0.0	1203
	(0.5)	(0.4)	

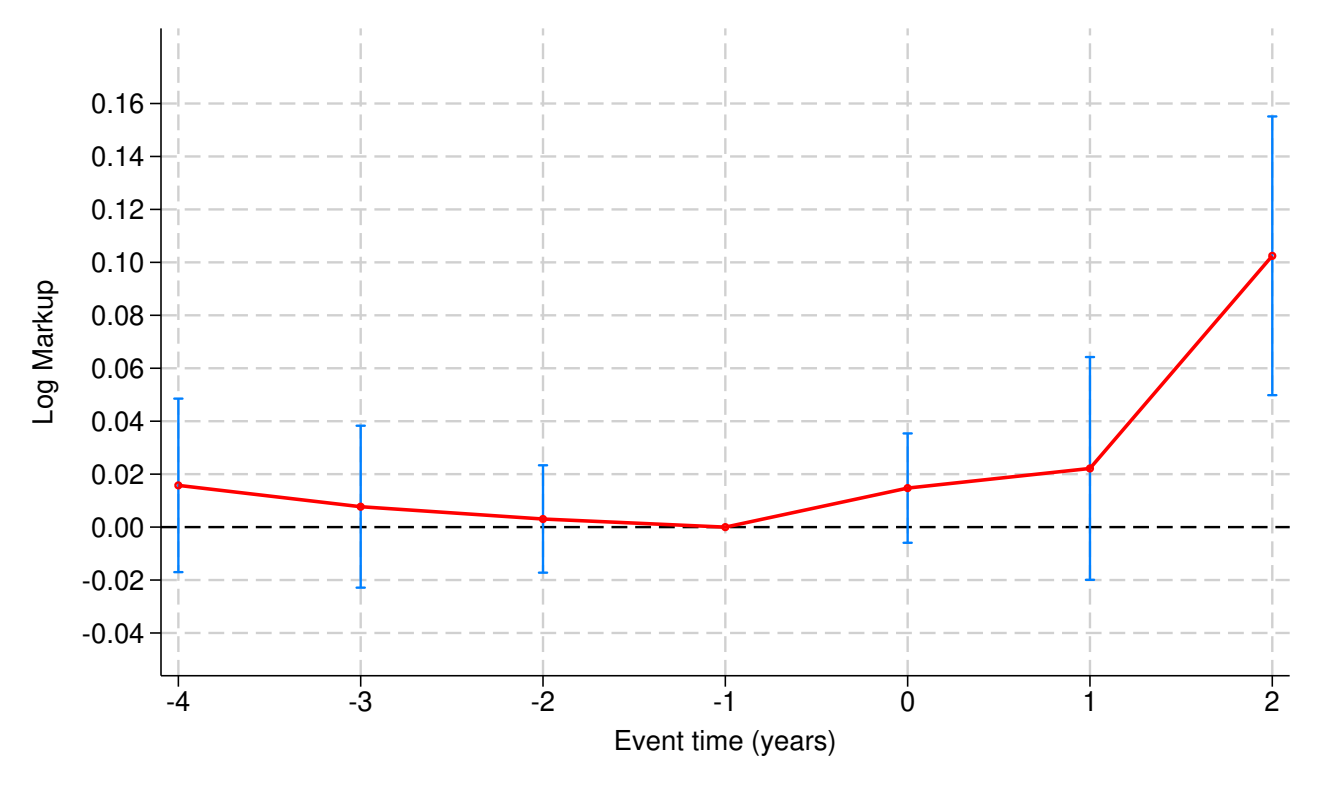
Testing Market Power: Robustness Checks

Back

- Different FEs: sector + country + year, sector + year + country × year Results
- Different Non-Absorbing options: restricting control group to not-yet treated only, restricting treated group to first time entries only, both combined. Results
- Different value of stabilisation effect parameter L: $L = 2$. Results
- Control for autoregressive patterns: control for 1 lag of y variable. Results

Markups — Robustness: Different FEs (sector + country + year)

Back

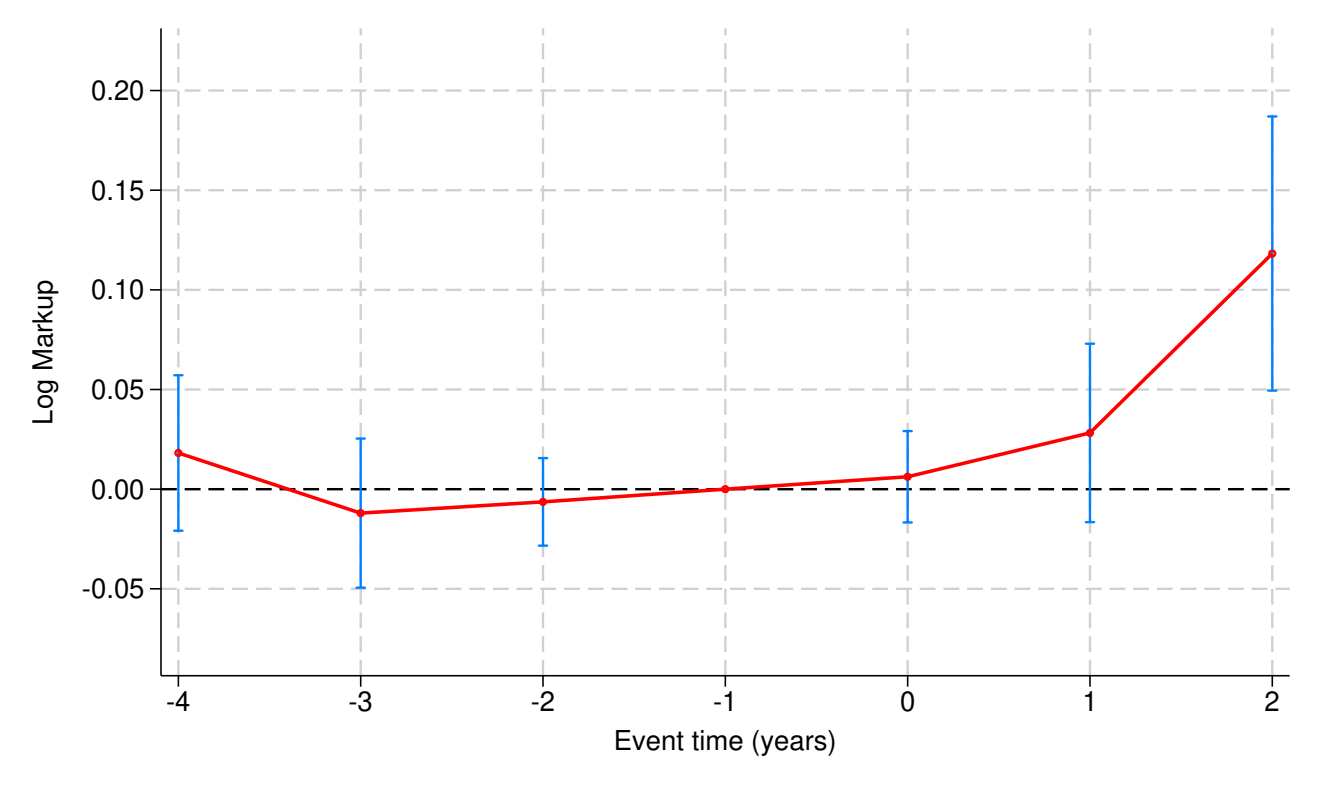


Pooled Estimates

	Estimate	SE	p-value	90% CI		Obs
Pre	0.0116976	0.0156784	0.4569	-0.0142733	0.0376685	646
Post	**0.0650473**	0.0258637	0.0131	0.0222177	0.1078769	716

Markups — Robustness: Different FEs (sector + country × year)

Back

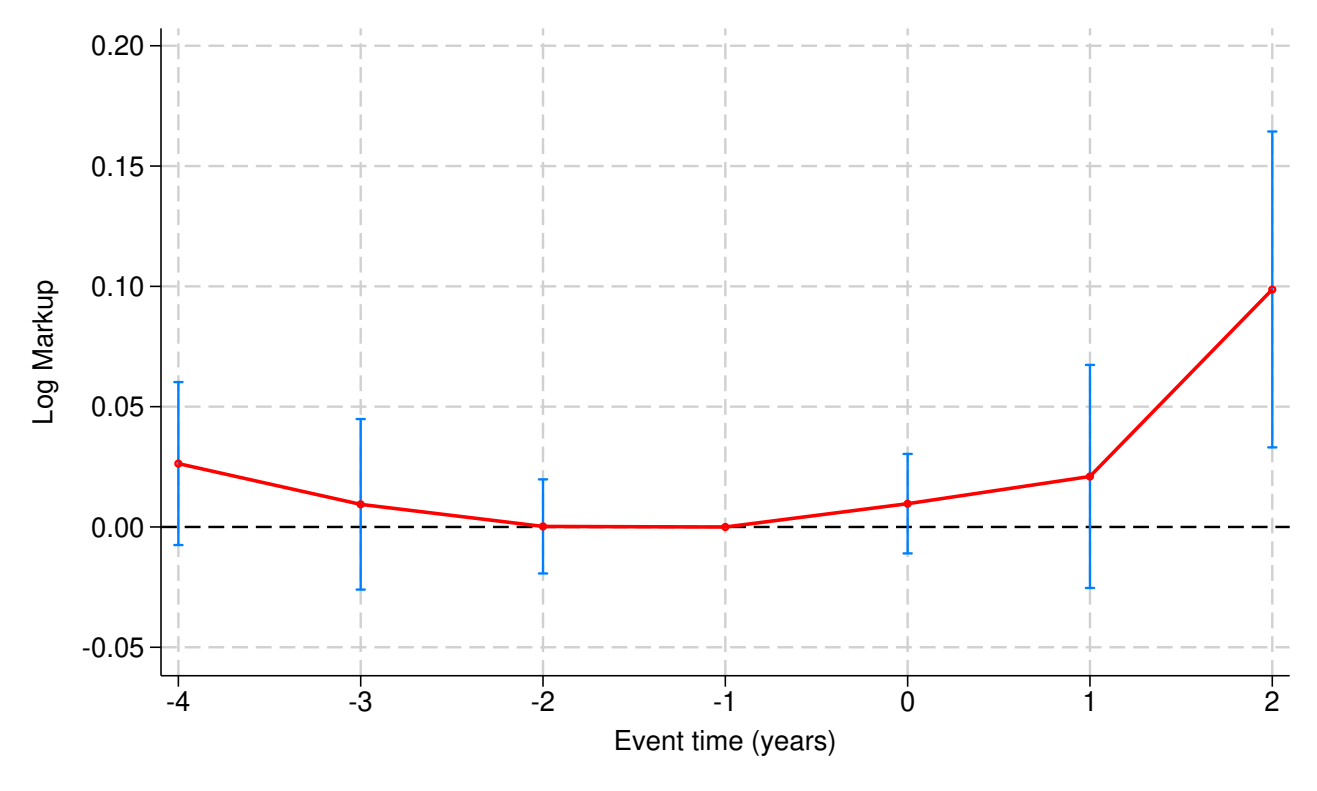


Pooled Estimates

	Estimate	SE	p-value	90% CI		Obs
Pre	0.0049123	0.0190503	0.7969	-0.0266511	0.0364756	605
Post	**0.1002327**	0.0386869	0.0106	0.0361521	0.1643133	669

Markups — Robustness: Non-Abs (controls = not-yet-treated)

Back

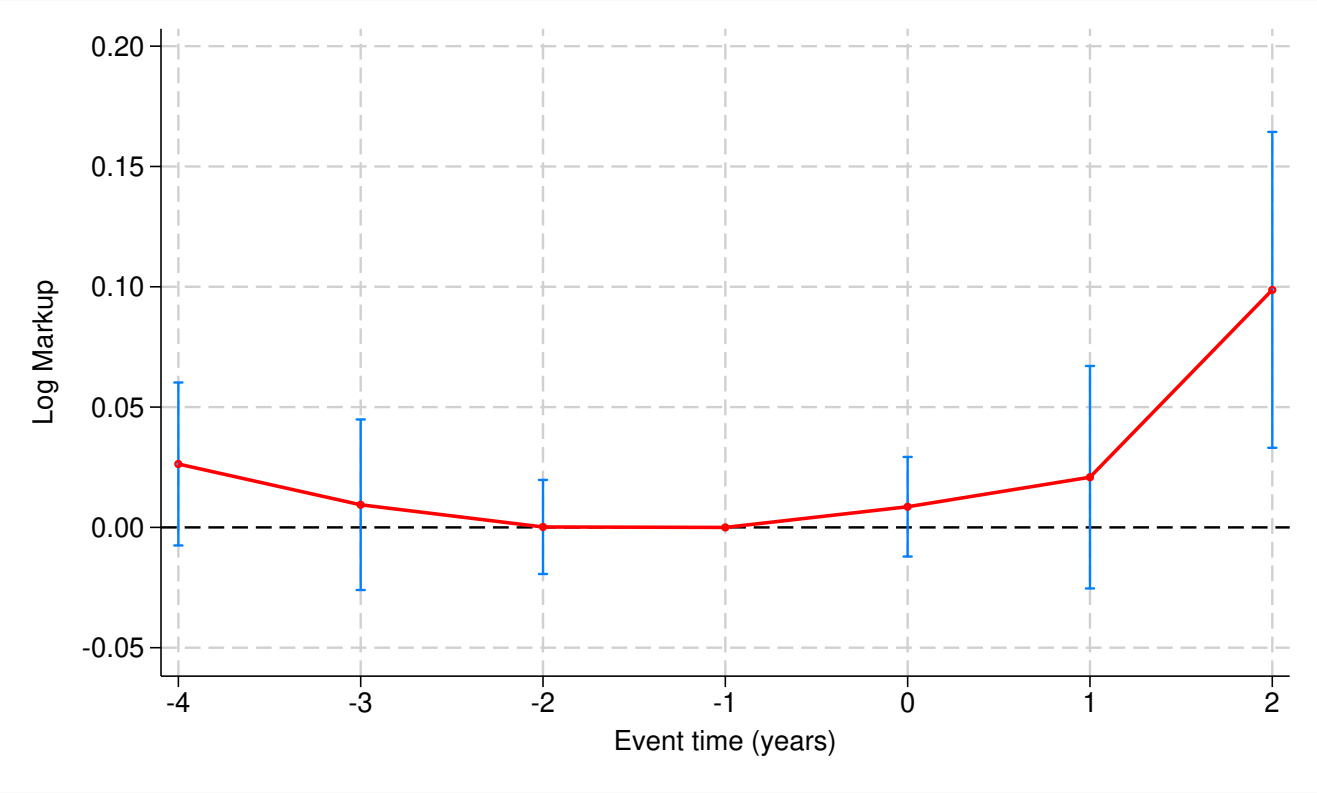


Pooled Estimates

	Estimate	SE	p-value	90% CI		Obs
Pre	0.0132584	0.0142507	0.3540	-0.0103627	0.0368796	536
Post	**0.0693607**	0.0298355	0.0217	0.0199278	0.1187935	588

Markups — Robustness: Non-Abs (treated = first-time)

Back

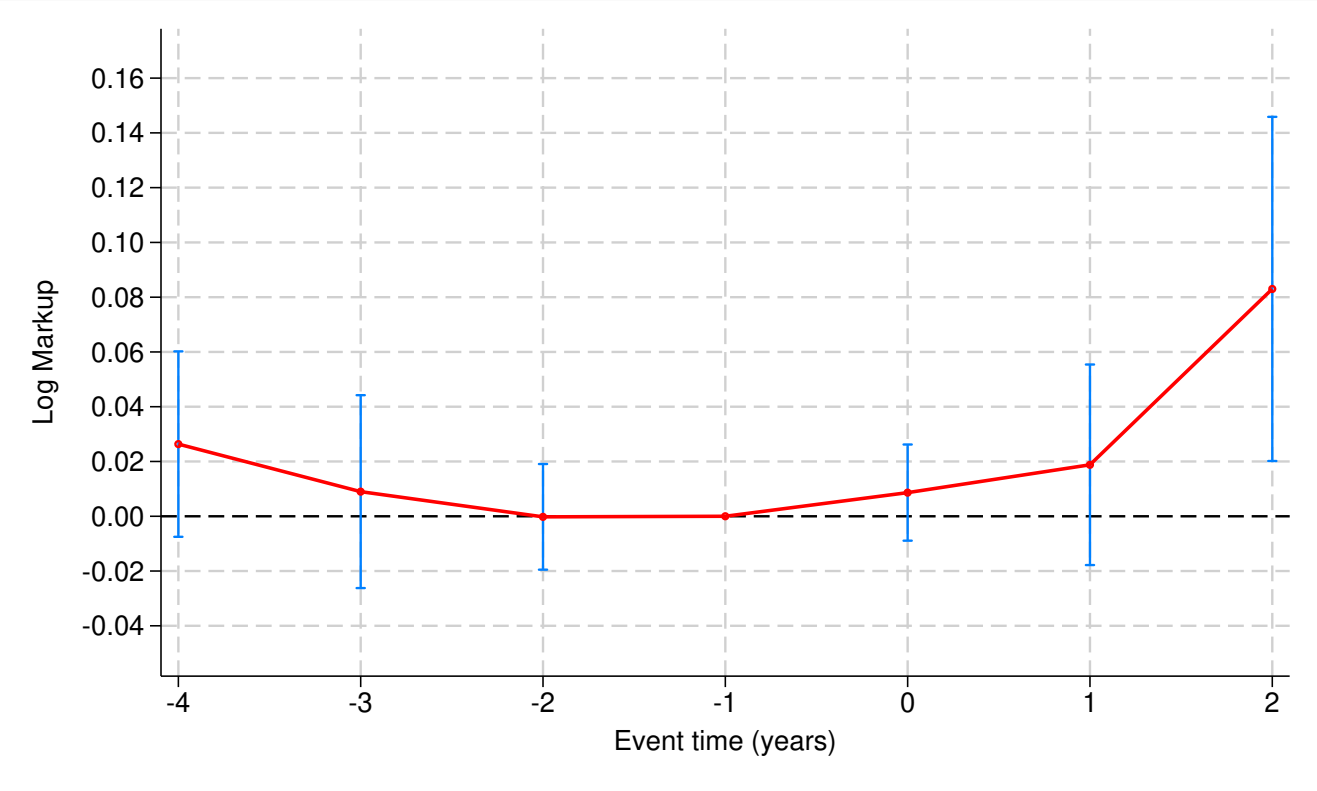


Pooled Estimates

	Estimate	SE	p-value	90% CI		Obs
Pre	0.0132584	0.0142507	0.3540	-0.0103627	0.0368796	536
Post	**0.0693607**	0.0298355	0.0217	0.0199278	0.1187935	588

Markups — Robustness: Non-Abs (both combined)

Back

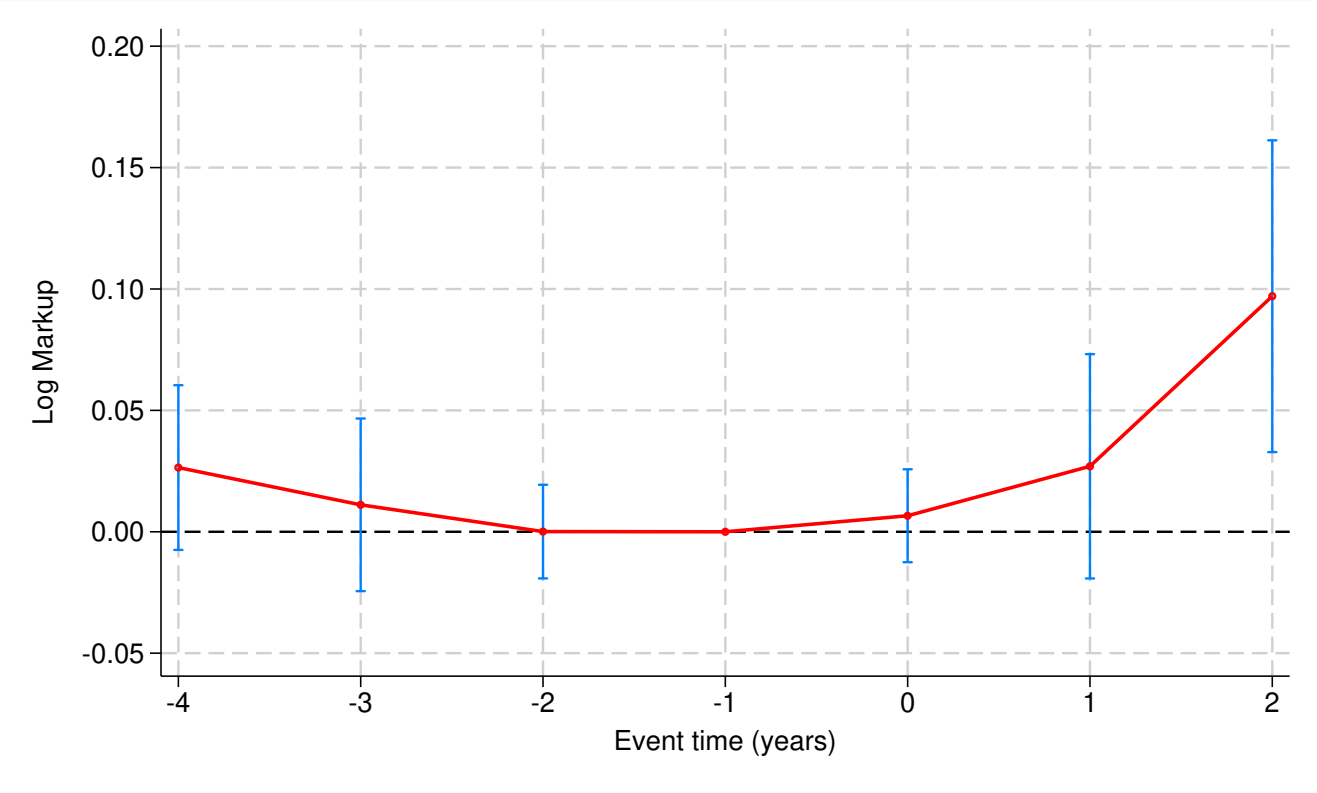


Pooled Estimates

	Estimate	SE	p-value	90% CI		Obs
Pre	0.0132584	0.0142507	0.3540	-0.0103627	0.0368796	536
Post	**0.0622688**	0.0277308	0.0264	0.0163258	0.1082118	589

Markups — Robustness: Stabilisation parameter $L = 2$

Back

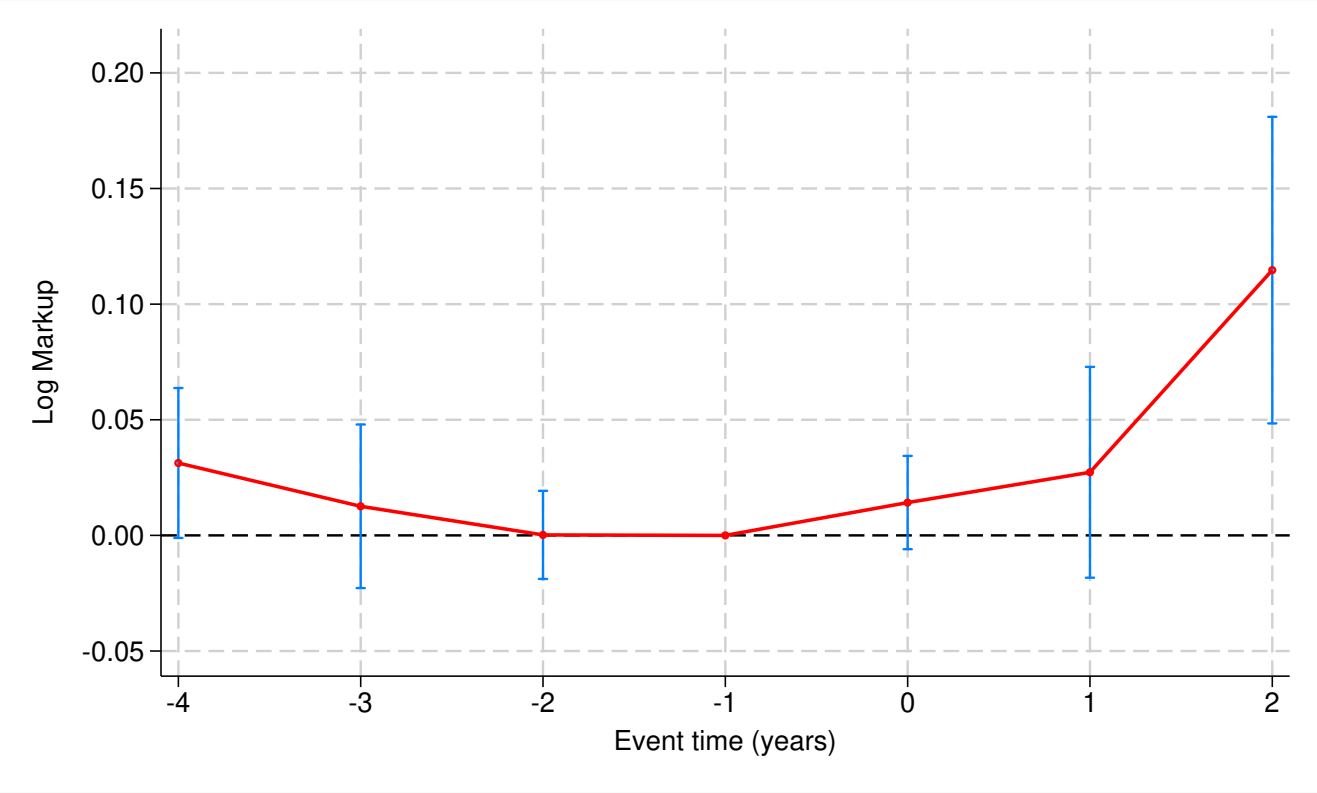


Pooled Estimates

	Estimate	SE	p-value	90% CI		Obs
Pre	0.0138530	0.0143636	0.3367	-0.0099553	0.0376612	540
Post	**0.0689886**	0.0291597	0.0195	0.0206810	0.1172962	603

Markups — Robustness: AR control (1 lag of y)

Back



Pooled Estimates

	Estimate	SE	p-value	90% CI		Obs
Pre	0.0152542	0.0134056	0.2574	-0.0069662	0.0374746	536
Post	**0.0820853**	0.0302972	0.0077	0.0318875	0.1322832	592

Model — Aggregates

Back

Aggregate Resource Constraints:

$$\begin{aligned}
\text{date } 0: &\quad E + \int_j \sum_i e(i)\, dj = C_0 + K_0, \\
\text{date } 1: &\quad Y_1 = C_1 + K_1 - (1-\delta)K_0 + h_0^B + h_0^{CB}, \\
\text{date } 2: &\quad Y_2 + (1-\delta)K_1 = C_2 + h_1^B + h_1^{CB}.
\end{aligned}$$

Welfare: reported in differences, free of the primitives

$$\begin{aligned}
\Delta\mathcal{W} = -\Delta K_0 &+ \beta\Big[\Delta Y_1 - \Delta K_1 + (1-\delta)\Delta K_0 - \Delta h_0^B - \Delta h_0^{CB} - \Delta v(N_1)\Big] \\
&+ \beta^2\Big[\Delta Y_2 + (1-\delta)\Delta K_1 - \Delta h_1^B - \Delta h_1^{CB} - \Delta v(N_2)\Big]
\end{aligned}$$

with $v(N) \equiv N^{1+1/\varphi}/(1+1/\varphi)$.

Model — Mapping Bond Prices and Interest Rates (1)

Back

One-period bonds

Gross return:

$$1 + r_1 = \frac{\text{cash at } t = 2 + \text{value after payment}}{\text{price at } t = 1} = \frac{1 + 0}{Q_1} = \frac{1}{Q_1}$$

$$\Rightarrow r_1 = \frac{1}{Q_1} - 1$$

Decomposition of payment

$$\underbrace{1}_{\text{total payment}} = \underbrace{(Q_1 - 0)}_{\text{principal repaid}} + \underbrace{r_1 Q_1}_{\text{interest payment}}$$

Model — Mapping Bond Prices and Interest Rates (2)

Back

Two-period bonds

Gross return:

$$1 + r_0 = \frac{\text{cash at } t = 1 + \text{value after payment}}{\text{price at } t = 0} = \frac{(1 - \lambda) + \lambda Q_1}{Q_0}$$

$$\Rightarrow r_0 = \frac{(1 - \lambda) + \lambda Q_1}{Q_0} - 1$$

Decomposition of payment (t=1)

$$\underbrace{1 - \lambda}_{\text{total payment}} = \underbrace{(Q_0 - \lambda Q_1)}_{\text{principal repaid}} + \underbrace{r_0 Q_0}_{\text{interest payment}}$$

Decomposition of payment (t=2)

$$\underbrace{\lambda}_{\text{total payment}} = \underbrace{(\lambda Q_1 - 0)}_{\text{principal repaid}} + \underbrace{\lambda(1 - Q_1)}_{\text{interest payment}}$$

$$\lambda = \lambda Q_1 + \lambda Q_1 r_1$$

Equilibrium in Goods Market

Back

$$\mu_{t+1,j}(i) = \frac{p_{t+1,j}(i)}{\mathrm{MC}_{t+1,j}(i)} = \frac{\varepsilon_{t+1,j}(i)}{1+\varepsilon_{t+1,j}(i)}$$

$$\frac{1}{\varepsilon_{t+1,j}(i)} = \frac{\partial p_{t+1,j}(i)}{\partial y_{t+1,j}(i)} \frac{y_{t+1,j}(i)}{p_{t+1,j}(i)} = -\frac{1}{\eta}\Big(1 - s_{t+1,j}(i)\Big) - \frac{1}{\theta} s_{t+1,j}(i)$$

$$s_{t+1,j}(i) = \frac{p_{t+1,j}(i) y_{t+1,j}(i)}{\sum_{i=1}^{I} p_{t+1,j}(i) y_{t+1,j}(i)} = \frac{\big(y_{t+1,j}(i)\big)^{1-\frac{1}{\eta}}}{\sum_{i=1}^{I} \big(y_{t+1,j}(i)\big)^{1-\frac{1}{\eta}}}$$

Analytical Model — Full Intermediate Firms Problem

Back

Intermediate firms - Problem at $t = 0$:

$$V_0 = \max_{a_0} \{M_1 V_1(z_1, k_0, a_0, y_1(-i))\}$$

$$\text{s.t. } k_0 = e + Q_0 a_0$$

Intermediate firms - Problem at $t = 1$:

$$V_1(z_1, k_0, a_0, y_1(-i)) = \max_{div_1, k_1, a_1} \{div_1 + M_2 V_2(z_2, k_1, a_0, a_1, y_2(-i))\}$$

$$\text{s.t. } 0 \leq div_1 \leq \Upsilon_1 - (1-\lambda)a_0 + (1-\delta)k_0 - k_1 + Q_1 a_1$$

Intermediate firms - Problem at $t = 2$:

$$V_2(z_2, k_1, a_0, a_1, y_2(-i)) = \max_{div_2} \{div_2\}$$

$$\text{s.t. } 0 \leq div_2 \leq \Upsilon_2 - (a_1 + \lambda a_0) + (1-\delta)k_1$$

Bond Mapping Quantitative model

Analytical Model — Intermediate Firms at $t = 1$

Back

Lagrangian:

$$\mathcal{L} = div_1 + M_2\Big[\Upsilon_2 - (a_1 + \lambda a_0) + (1-\delta)k_1\Big] + \nu_1\, div_1$$
$$+ \vartheta_1\Big[\Upsilon_1 - (1-\lambda)a_0 + (1-\delta)k_0 - k_1 + Q_1 a_1 - div_1\Big]$$

First-order conditions:

$$[div_1]:\ 1 + \nu_1 = \vartheta_1, \qquad [a_1]:\ \vartheta_1 Q_1 = M_2, \qquad [k_1]:\ \vartheta_1 = M_2\Big[\frac{\partial \Upsilon_2}{\partial k_1} + 1 - \delta\Big]$$

$\Rightarrow$ for every firm: $\dfrac{\partial \Upsilon_2}{\partial k_1} = \Big(\dfrac{1}{Q_1} - 1\Big) + \delta = r_1 + \delta$

Solution:

1. **Unconstrained firms** ($div_1 > 0 \Rightarrow \nu_1 = 0,\ \vartheta_1 = 1$):
   - $Q_1 = M_2 = \beta$: bond supply perfectly elastic at the frictionless price; div_1 is the budget residual
2. **Constrained firms** ($div_1 = 0 \Rightarrow \nu_1 \geq 0,\ \vartheta_1 \geq 1$):
   - Q_1 from the intermediary's FOC;
     $a_1 = \big[k_1^* - \Upsilon_1 + (1-\lambda)a_0 - (1-\delta)k_0\big]/Q_1$ from the budget

Analytical Model — Intermediate Firms at $t = 0$

Back

Problem:

$$\max_{a_0} \; M_1 V_1(a_0) \qquad \text{s.t.} \quad k_0 = e + Q_0 a_0, \quad \text{FOCs at } t = 1$$

Envelope: with internal funds $N_1 \equiv \Upsilon_1 - (1-\lambda)a_0 + (1-\delta)k_0$, $\partial V_1 / \partial N_1 = \vartheta_1$ for every firm

First-order condition:

$$\vartheta_1 \Big[Q_0 \Big(\frac{\partial \Upsilon_1}{\partial k_0} + 1 - \delta \Big) - (1-\lambda) \Big] = M_2 \, \lambda$$

Solution: using $[a_1]$, $\vartheta_1 = M_2 / Q_1$, the multiplier divides out: **same condition for both types**

$$\frac{\partial \Upsilon_1}{\partial k_0} = \frac{(1-\lambda) + \lambda Q_1}{Q_0} - 1 + \delta = r_0 + \delta, \qquad a_0 = \frac{k_0^* - e}{Q_0}$$

$\Rightarrow$ net worth e moves only the level of issuance, not the optimality condition

Analytical Model — Private Financial Intermediary

Back

Problem at $t = 1$:

$$\max_{a_1^B, d_1} V_1^B = \int_j \sum_{i=1}^{I} (M_2 - Q_1)\, a_1^B\, dj + \left(M_2 - Q_1^d\right) d_1 - M_2 \int_j \sum_{i=1}^{I} \frac{\gamma_1}{2} \frac{\left(Q_1 a_1^B\right)^2 + \left(\lambda Q_0 a_0^B\right)^2}{k_0}\, dj$$

- $[a_1^B]: \quad Q_1 = M_2 \left[1 - \gamma_1 Q_1 \frac{Q_1 a_1^B}{k_0}\right] \qquad [d_1]: \quad Q_1^d = M_2 = \beta$

Problem at $t = 0$:

$$\max_{a_0^B, d_0} V_0^B = \int_j \sum_{i=1}^{I} \left[(1-\lambda) M_1 + \lambda M_1 M_2 - Q_0\right] a_0^B\, dj + \left(M_1 - Q_0^d\right) d_0$$
$$- M_1 \int_j \sum_{i=1}^{I} \frac{\gamma_0}{2} \frac{\left(Q_0 a_0^B\right)^2}{e}\, dj - M_1 M_2 \int_j \sum_{i=1}^{I} \frac{\gamma_1}{2} \frac{\left(\lambda Q_0 a_0^B\right)^2}{k_0}\, dj$$

- $[a_0^B]: \quad Q_0 = (1-\lambda) M_1 + \lambda M_1 M_2 - M_1 Q_0 \left[\gamma_0 \frac{Q_0 a_0^B}{e} + M_2 \lambda \gamma_1 \frac{\lambda Q_0 a_0^B}{k_0}\right]$

  $[d_0]: \quad Q_0^d = M_1 = \beta$

$\Rightarrow$ **Segmented pricing:** $Q_t(i)$ depends only on firm i's own placed quantity and balance sheet

Quantitative Model — Full Intermediate Firms Problem

Back

Intermediate firms - Problem at $t = 0$:

$$V_0 = \max_{a_0} \{M_1 V_1(z_1, k_0, a_0, y_1(-i))\}$$

$$\text{s.t. } k_0 = e + Q_0 a_0, \qquad \log e(i) = a + b \log z_i + \zeta (\log z_i)^2$$

Intermediate firms - Problem at $t = 1$:

$$V_1 = \max_{div_1, k_1, a_1} \Big\{ div_1 + M_2 \Big[(1-\tau)\Upsilon_2 - (1 - \tau Q_1 r_1)(a_1 + \lambda a_0) + \big(1 - (1-\tau)\delta\big) k_1 \Big] \Big\}$$

$$\text{s.t. } 0 \leq div_1, \quad \varphi(div_1) \leq (1-\tau)\Upsilon_1 - a_0(1 - \lambda - \tau Q_0 r_0) + \big(1 - (1-\tau)\delta\big) k_0 - k_1 + Q_1 a_1$$

$$\varphi(div_1) = div_1 + \frac{\kappa_d}{k_0} \big(div_1 - \bar{d}(i)\big)^2 \qquad \text{(a transfer, rebated to the household)}$$

Intermediate firms - Problem at $t = 2$: no payout cost

$$div_2 = (1-\tau)\Upsilon_2 - (1 - \tau Q_1 r_1)(a_1 + \lambda a_0) + \big(1 - (1-\tau)\delta\big) k_1$$

Bond Mapping Analytical model

Quantitative Model — Intermediate Firms at $t = 1$

Back

Lagrangian: with $N_1 \equiv (1-\tau)\Upsilon_1 - a_0(1-\lambda-\tau Q_0 r_0) + \big(1-(1-\tau)\delta\big)k_0$

$$\mathcal{L} = div_1 + \beta\Big[(1-\tau)\Upsilon_2 - (1-\tau Q_1 r_1)(a_1 + \lambda a_0) + \big(1-(1-\tau)\delta\big)k_1\Big]$$
$$+ \vartheta_1\Big[N_1 + Q_1 a_1 - \varphi(div_1) - k_1\Big] + \nu_1\, div_1$$

First-order conditions:

$$[div_1]:\ 1 + \nu_1 = \vartheta_1\varphi'(div_1), \qquad [a_1]:\ \vartheta_1 = \beta\Big[\frac{1-\tau}{Q_1} + \tau\Big],$$
$$[k_1]:\ \vartheta_1 = \beta\Big[(1-\tau)\frac{\partial\Upsilon_2}{\partial k_1} + 1 - (1-\tau)\delta\Big]$$

Solution:

- **Capital:** $\dfrac{\partial\Upsilon_2}{\partial k_1} = \dfrac{1}{Q_1} - 1 + \delta = r_1 + \delta \quad \Rightarrow$ the tax cancels for every firm, at every value of ϑ_1
- **Payout schedule** (interior): $div_1 = \bar{d} + \dfrac{k_0}{2\kappa_d}\left(\dfrac{1}{\vartheta_1} - 1\right) \quad \Rightarrow \vartheta_1 > 1$
  while still paying: payout below target
- **Corner** $div_1 = 0$ when $\vartheta_1 \geq \big[1 - 2\kappa_d\bar{d}/k_0\big]^{-1}$

Quantitative Model — Intermediate Firms at $t = 0$

Back

Problem: $\max_{a_0} \beta V_1(a_0)$ s.t. $k_0 = e + Q_0 a_0$; envelope: $\partial V_1 / \partial N_1 = \vartheta_1$, whichever regime at $t = 1$

Four direct effects of a_0: internal funds through k_0; the maturing repayment; the date-2 tranche; the scale of the payout cost (k_0 in its denominator)

First-order condition:

$$\vartheta_1 \left[Q_0 \left((1-\tau) \frac{\partial \Upsilon_1}{\partial k_0} + 1 - (1-\tau)\delta + \frac{\kappa_d}{k_0^2} \left(div_1 - \bar{d} \right)^2 \right) - \left(1 - \lambda - \tau Q_0 r_0 \right) \right] = \beta \left(1 - \tau Q_1 r_1 \right) \lambda$$

Solution: using $[a_1]$, ϑ_1 divides out

$$\frac{\partial \Upsilon_1}{\partial k_0} = r_0 + \delta - \frac{\kappa_d \left(div_1 - \bar{d} \right)^2}{(1-\tau)\, k_0^2}, \qquad a_0 = \frac{k_0^* - e}{Q_0}$$

$\Rightarrow$ **Scale-relief term:** second order in the payout deviation; zero at the no-policy point ($div_1 = \bar{d}$) and at $\kappa_d = 0$, where the analytical condition returns

Quantitative Model — Private Financial Intermediary

Back

Problem at $t = 1$: $\max_{a_1^B, d_1} \int_j \sum_i (\beta - Q_1)\, a_1^B\, dj + (\beta - Q_1^d)\, d_1 - \beta\, h_1^B$, with the one-sided h_1^B

- $[a_1^B]$: $\dfrac{1}{Q_1} - \dfrac{1}{\beta} = \gamma_1 \left[x_1 - \bar{x}_1\right]^+$: the spread is γ_1 times the excess leverage, zero on the flat segment
- Flat segment: $Q_1 = \beta$; on the slope:
$$Q_1 = \frac{2\beta}{(1 - \beta\gamma_1\bar{x}_1) + \sqrt{(1 - \beta\gamma_1\bar{x}_1)^2 + 4\beta^2\gamma_1 a_1^B / k_0}}$$

Problem at $t = 0$: both holding periods of the multi-period bond, each on the excess over its reference

- $[a_0^B]$: $Q_0 = \bar{\beta} - \Gamma_0\, Q_0 \left[Q_0 a_0^B - \bar{x}_0 e\right]^+$, $\bar{\beta} \equiv \beta\left[(1 - \lambda) + \beta\lambda\right]$,
$\Gamma_0(i) \equiv \dfrac{\beta\gamma_0}{e(i)} + \dfrac{\beta^2\gamma_0\lambda^2}{k_0(i)}$
- Flat segment: $Q_0 = \bar{\beta}$; on the slope:
$$Q_0 = \frac{2\bar{\beta}}{(1 - \Gamma_0\bar{x}_0 e) + \sqrt{(1 - \Gamma_0\bar{x}_0 e)^2 + 4\bar{\beta}\,\Gamma_0\, a_0^B}}; \quad [d_t] : \; Q_t^d = \beta$$

Regime test: on the slope iff the placement at the safe price exceeds the reference holding; on the flat segment purchases move nothing real

Markup Response: a Structural Lower Bound

Back

The gap:

- Purchased minus never-purchased firms: 0.2 log points in the model vs 6.9 [2.0, 11.9] in the data $\Rightarrow$ ten times below the lower end of the interval
- Relative to the capital response: 0.066 in the model vs about 0.7 in the data

Not measurement:

- The production-approach markup overstates the true response if part of COGS is fixed; the COGS elasticity to revenue in the panel is 0.97 $\Rightarrow$ the measured markup tracks the true one

Market structure:

- Nested-CES Cournot (Atkeson and Burstein, 2008): the markup rises only through the market share, $\mu = \dfrac{\epsilon}{\epsilon - 1}, \quad \dfrac{1}{\epsilon} = \dfrac{1}{\eta}(1 - s) + \dfrac{1}{\theta} s$
- Elasticity of the markup to the firm's own capital capped at $\alpha/(1-\alpha) = 0.5 \Rightarrow$ no parameter value matches the markup response for the capital response the financing block delivers

$\Rightarrow$ **Every markup effect of the quantitative model is a lower bound on the market-power cost of the programme**

Quantitative Model — Simpler Economies

Back

Economy	Safe rate	First-issue yield effect (pp)	Capital response	Payout response	Purchased firms at a payout limit
Price only (analytical economy)	4%	−0.216	0.000	–	0 of 70
Payout floor	4%	−0.216	0.000	–	0 of 70
Payout cost, no reference holding	4%	−0.216	0.012	0.0098	0 of 70
This paper	2%	−0.217	0.031	0.0055	–
			0.099	−0.012	
Data		−0.217	[0.028, 0.171]	[−0.030, 0.0055]	

Some Facts about the CSPP Data

Back

	Issues	Purchased	Share
Issuance-week purchase probability, by the issuer's previous issue			
First eligible issue of the issuer	340	195	**0.574**
Previous issue purchased in its issuance week, any gap	957	727	0.760
within 1 year	762	627	**0.823**
within 90 days	495	447	0.903
Previous issue not purchased in its issuance week, any gap	955	161	0.169
within 1 year	843	126	**0.149**
within 90 days	531	55	0.104
Last two issues both purchased in their issuance week	637	493	0.774

Calibration — Other Externally Set Parameters

Back

Parameter	Value	Description	Source
Preferences and technology			
β	0.9804	Discount factor	2% safe rate
α	0.333	Capital share	Standard
δ	0.08	Depreciation rate	Data
η	5.75	Within-sector elasticity of substitution	De Loecker et al. (2021)
θ	1.2	Across-sector elasticity of substitution	De Loecker et al. (2021)
φ	0.25	Frisch elasticity of labour supply	Chetty et al. (2011)
$I \times J$	12×44	Firms per sector $\times$ sectors	Data
σ_z	0.35	Dispersion of log productivity	Data
ρ_z	0.959	Persistence of log productivity	Data
Firms and finance			
τ	0.246	Corporate tax rate	Data
λ	0.75	Share of the first bond repaid at date 2	Jeenas (2025), annualised
κ	0.001	Central bank's intermediation cost	Gertler and Karadi (2011)
$\phi^{\max}$	0.70	Per-issue purchase cap	ECB legal rule
E	25	Household endowment at date 0	Normalisation

Calibration — Parameters Fitted inside Every Solve

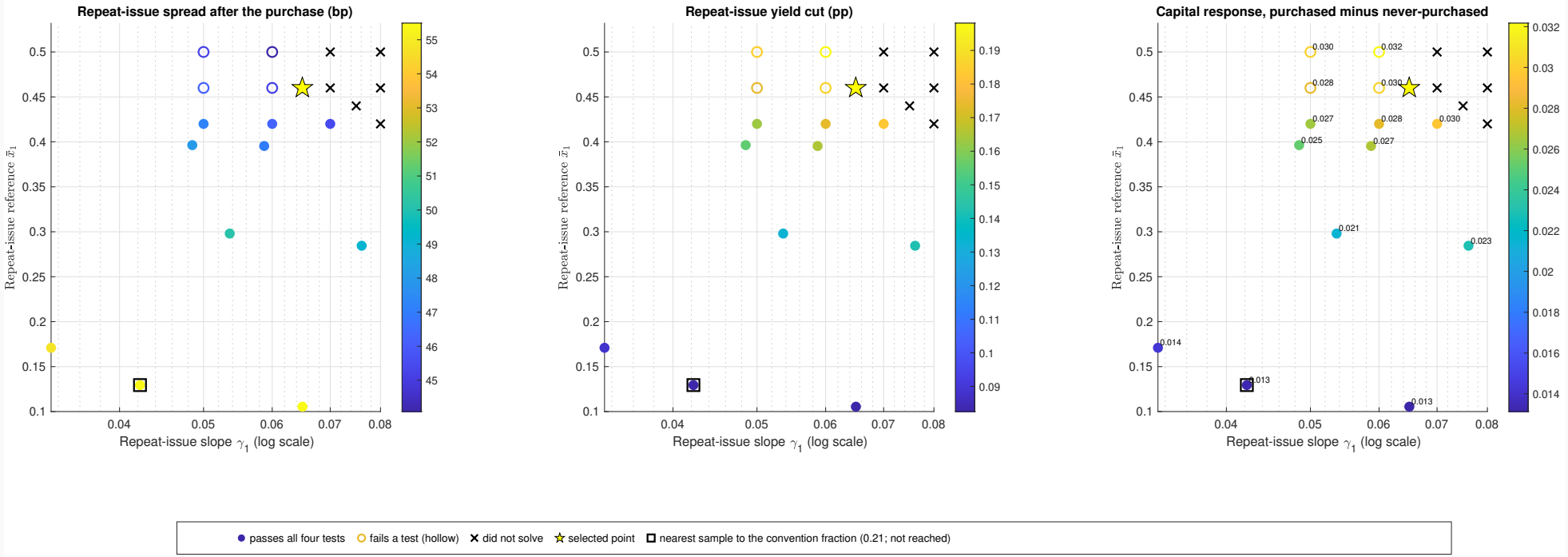
Back

Parameter	Value	Description	Moment	Model	Data
B. Fitted inside every solve					
ψ_1	3.60e-04	Overhead labour, scale	Overhead cost over revenue	0.283	0.283
ψ_2	2.920	Overhead labour, curvature in type	Overhead cost elasticity to size	0.887	0.887
a	-6.616	Net worth, level	Net worth over assets, mean	0.357	0.357
b	3.571	Net worth, slope in log type	Net worth over assets, bottom size decile	0.352	0.352
ζ	-1.632	Net worth, curvature in log type	Net worth over assets, top size decile	0.364	0.364

Calibration — Repeat-Issue Pair: Parameters Explored

Back

Repeat-issue parameters explored, safe rate 2.0 % (first issue held at $\gamma_0 = 0.0079$, $\bar{x}_0 = 0.32$): 21 runs, 6 did not solve; the search ends at the solver wall



Calibration at Other Safe Rates and at the Alternative Anchor

Back

	First-issue yield effect (pp)	First-issue spread (bp)	Repeat-issue yield effect (pp)	Repeat-issue spread (bp)	Capital response	Payout response	Consumption, date 1	K_1/K_0	What ends the search
Safe rate 0.5%	−0.217	119.4	−0.087	13.7	0.0165	0.00086	0.821	1.093	the spread floor and the solver wall
Safe rate 2.0% (headline)	−0.217	119.9	−0.191	44.7	0.0310	0.00546	0.940	1.053	the solver wall
Safe rate 3.0%	−0.216	119.7	−0.213	75.3	0.0308	0.00531	1.129	0.973	the payout ceiling and the solver wall
Safe rate 4.0%	−0.217	120.0	−0.219	107.3	0.0286	0.00528	1.243	0.909	the payout ceiling
Safe rate 2.0%, first-issue anchor −0.266	−0.267	120.3	−0.192	43.7	0.0311	0.00546	0.924	1.058	first issue re-pinned; repeat issue held at the headline point

The capital response is about 1.71 × the proportional fall in purchased firms' cost of capital at every rate. From 3 to 4% the rate cut is held at the target while the cost of capital keeps rising.

Quantitative Model — Pricing and Payout Regimes

Back

	Flat, pays dividends	Flat, no dividend	Sloped, pays dividends	Sloped, no dividend	Flat at date 0	Firms
No policy, all	0	0	416	112	0	528
No policy, purchased	0	0	67	3	0	70
No policy, never	0	0	47	5	0	52
Announced benchmark, all	0	0	419	109	0	528
Announced benchmark, purchased	0	0	67	3	0	70
Announced benchmark, never	0	0	49	3	0	52
Realised path ($\rho = 0.823$), all	0	0	420	108	0	528
Realised path ($\rho = 0.823$), purchased	0	0	69	1	0	70
Realised path ($\rho = 0.823$), never	0	0	47	5	0	52

Quantitative Model — Persistence of the Repeat Purchase

Back

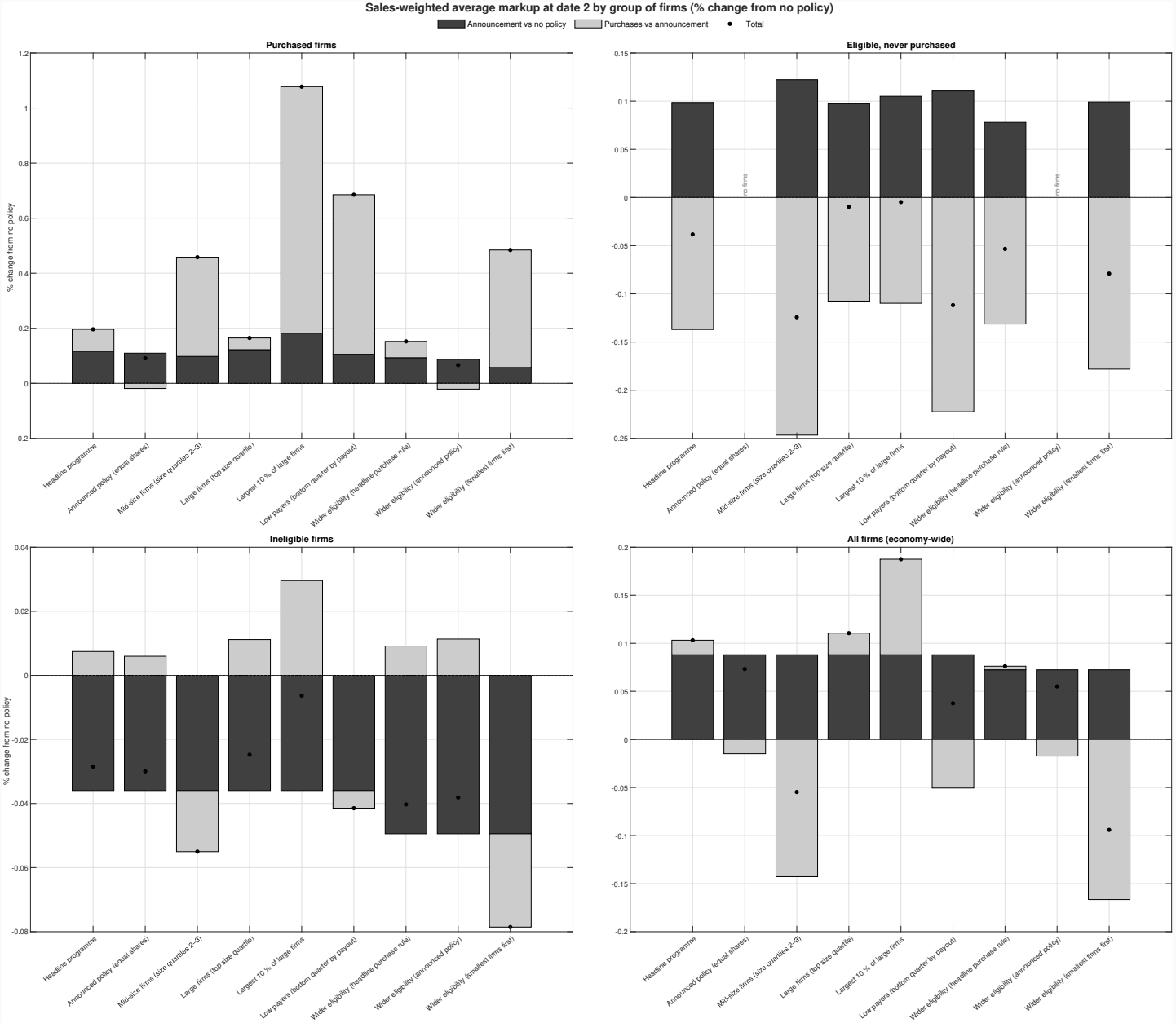
	Repeat-issue yield effect (pp)	Repeat-issue spread (bp)	Capital response	Payout gap, pre-purchase capital	Markup gap (log)
Repurchase probability 0.823	-0.1913	44.7	+0.0310	+0.0055	+0.00205
Repurchase probability 0.760	-0.1760	46.2	+0.0286	+0.0050	+0.00187
Repurchase probability 0.903	-0.2108	42.8	+0.0341	+0.0060	+0.00229
Repurchase probability 0.000	-0.0003	64.1	+0.0009	-0.0001	-0.00021
Repurchase probability 1.000	-0.2348	40.3	+0.0380	+0.0068	+0.00258

Markup Change by Group, Headline Programme

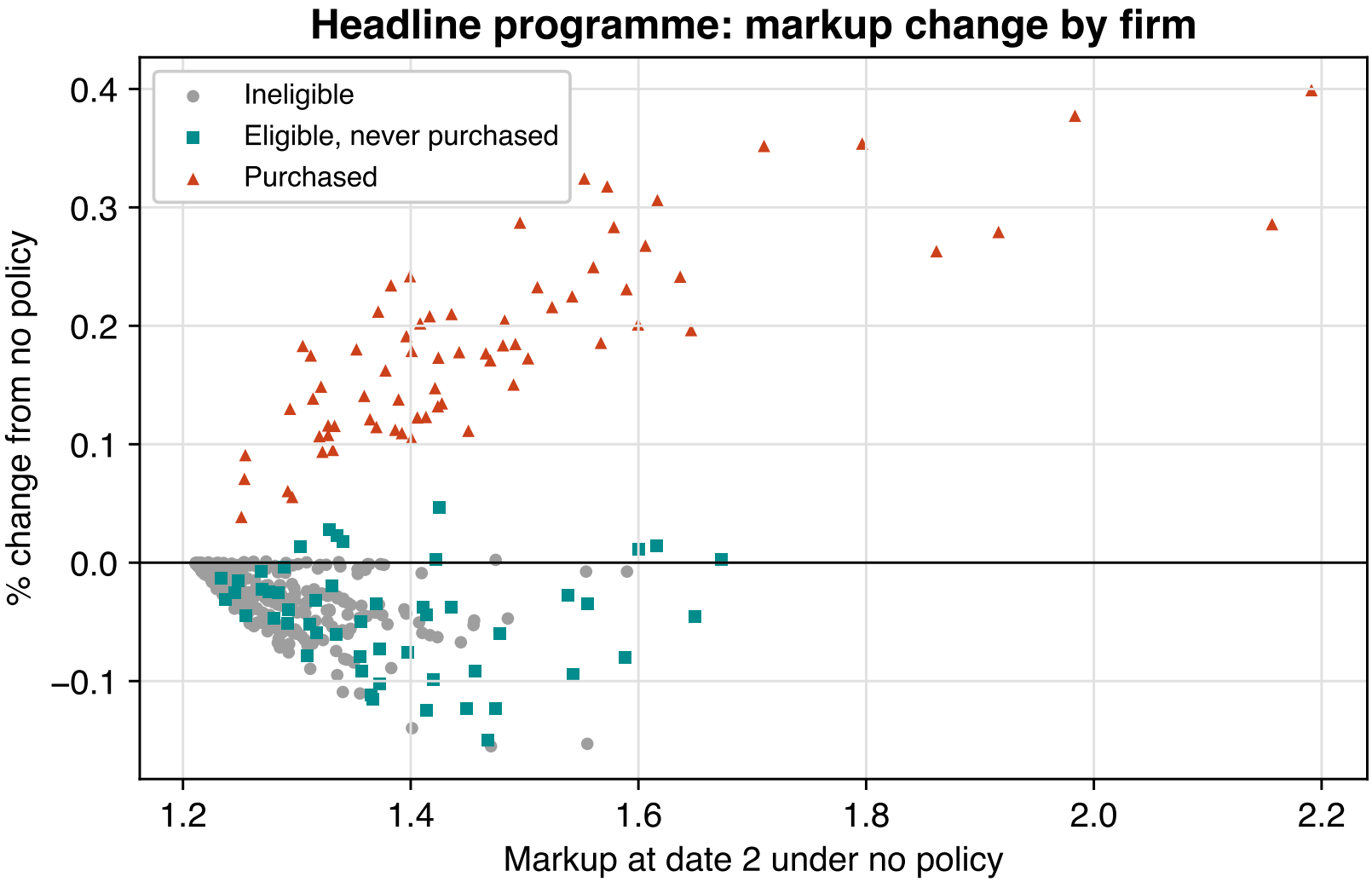
	Announcement part	Purchase part	Total	Single purchase
Purchased	+0.1170	+0.0794	+0.1964	+0.0117
Eligible, never purchased	+0.0986	−0.1369	−0.0383	+0.0065
Ineligible	−0.0359	+0.0074	−0.0285	−0.0033
All firms (economy-wide)	+0.0881	+0.0151	+0.1032	+0.0084

Markup Change by Group, All Designs

Back

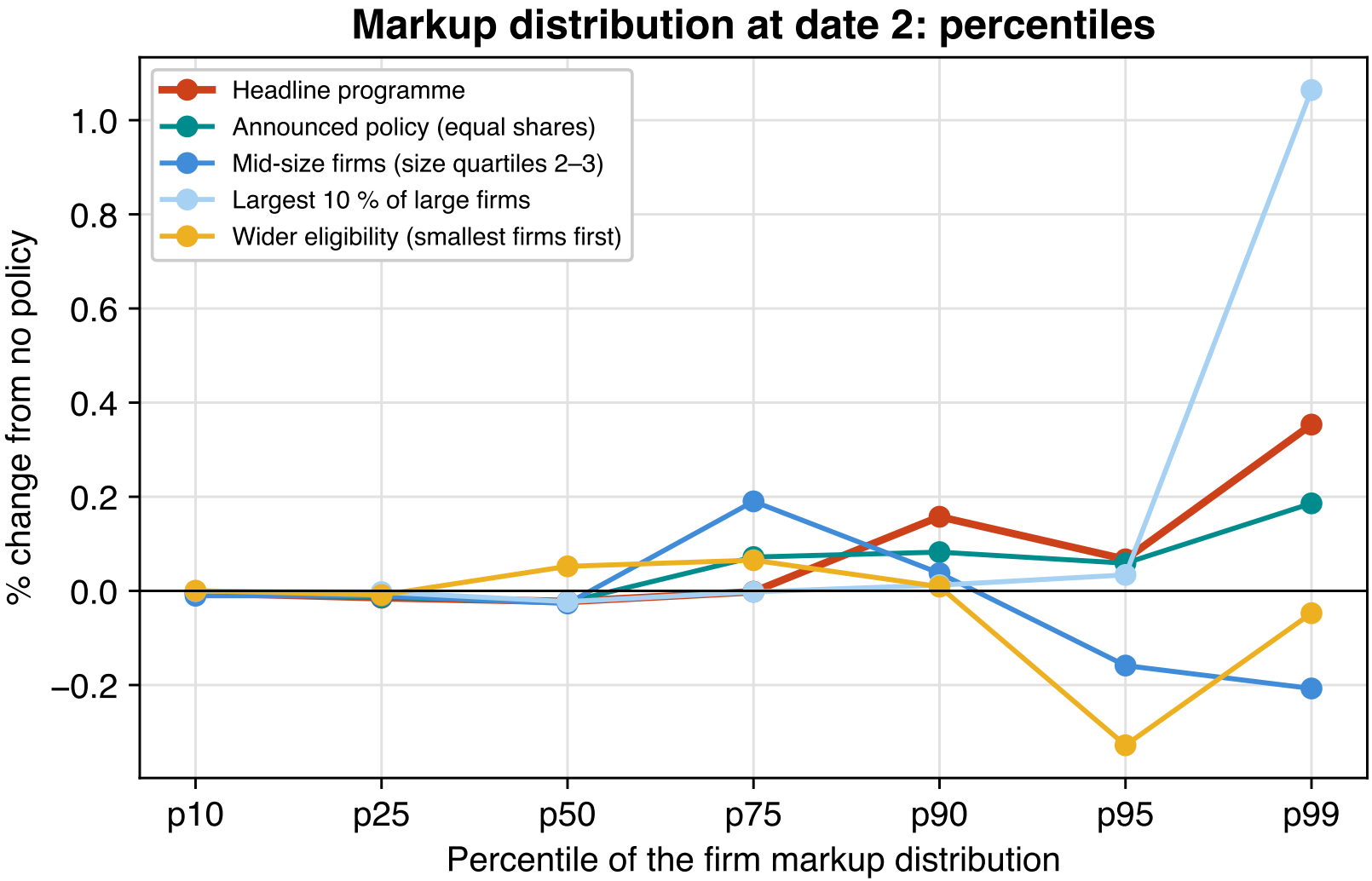


Markup Change by Firm, Headline Programme



Markup Distribution: Percentiles by Design

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Purchase Designs at the Headline's First-Issue Euros

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Design	Firms bought	Share bought per issue	Euros, both dates (headline = 1)	Capital gained per euro	Output, date 2 (%)	Markup, date 2 (%)	Welfare (%)	Welfare per euro
Headline programme	70	0.131	1.000	0.205	0.437	0.103	0.0124	0.0240
Announced policy (equal shares)	122	0.085	0.993	0.206	0.427	0.073	0.0124	0.0241
Mid-size firms (size quartiles 2–3)	46	0.373	1.032	0.238	0.449	−0.055	0.0084	0.0158
Large firms (top size quartile)	76	0.111	1.006	0.205	0.443	0.111	0.0126	0.0243
Largest 10% of large firms	8	0.700	1.041	0.117	0.303	0.188	0.0058	0.0108
Low payers (bottom quarter by payout)	31	0.433	1.064	0.254	0.539	0.038	0.0102	0.0184
Wider eligibility (headline purchase rule)	99	0.120	1.006	0.213	0.446	0.076	0.0126	0.0242
Wider eligibility (announced policy)	173	0.078	0.999	0.211	0.434	0.055	0.0125	0.0243
Wider eligibility (smallest firms first)	58	0.678	1.126	0.223	0.438	−0.097	0.0030	0.0052

Programme Size across Designs

Back

Design	Euros at the first issue (headline = 1)	Euros over both dates (headline = 1)
Headline programme	1.00	1.00
Announced policy (equal shares)	1.00	0.99
Mid-size firms (size quartiles 2–3)	1.00	1.03
Large firms (top size quartile)	1.00	1.01
Largest 10% of large firms	1.00	1.04
Low payers (bottom quarter by payout)	0.99	1.06
Wider eligibility (headline purchase rule)	1.01	1.01
Wider eligibility (announced policy)	1.01	1.00
Wider eligibility (smallest firms first)	1.01	1.13

First-issue euros equal by construction; the repeat issue follows the repurchase probability 0.823 and each design's repeat issuance.

Deviations from the Announced Policy

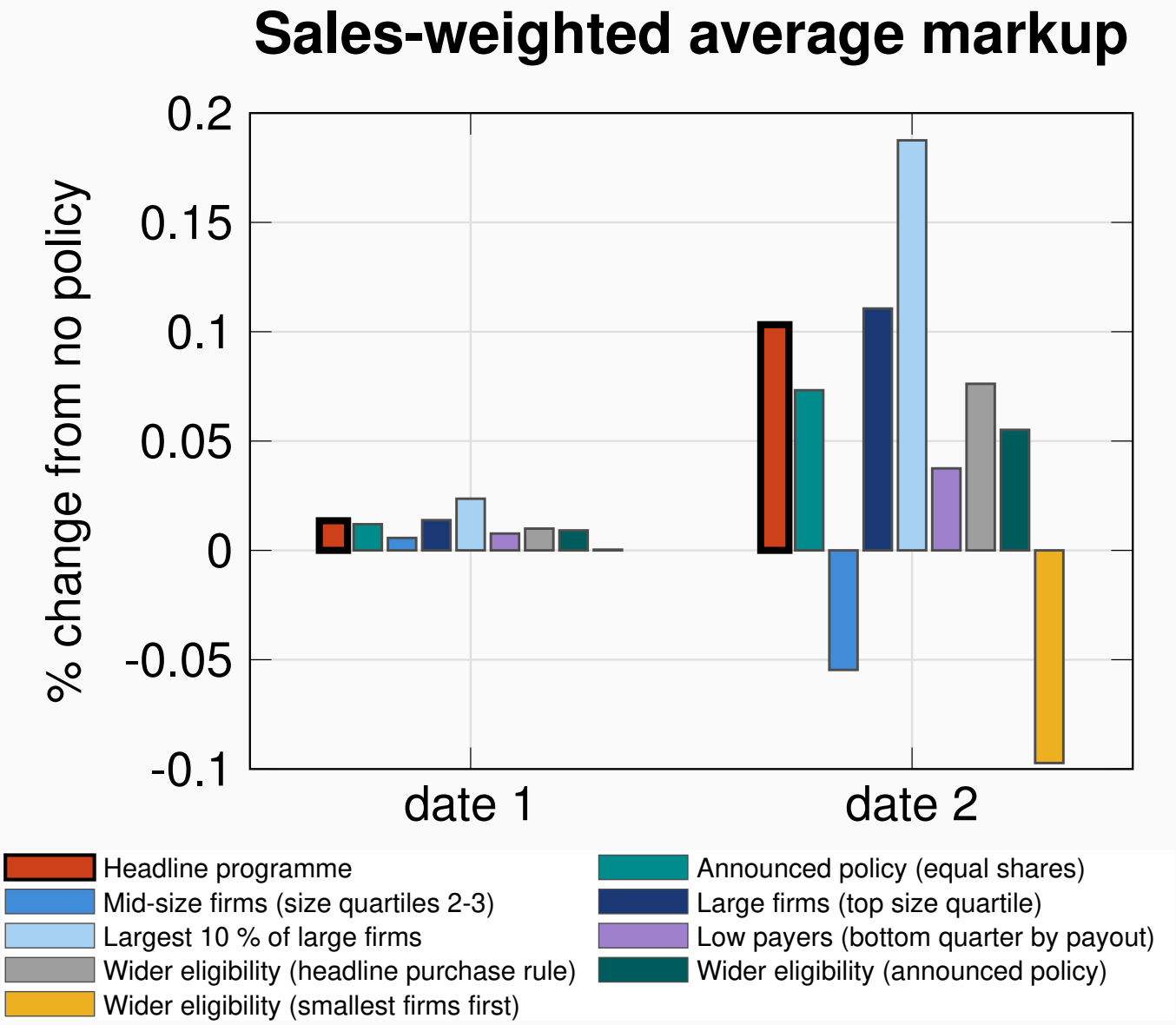
Back

- The announced benchmark buys **8.53%** of every eligible issue at both dates

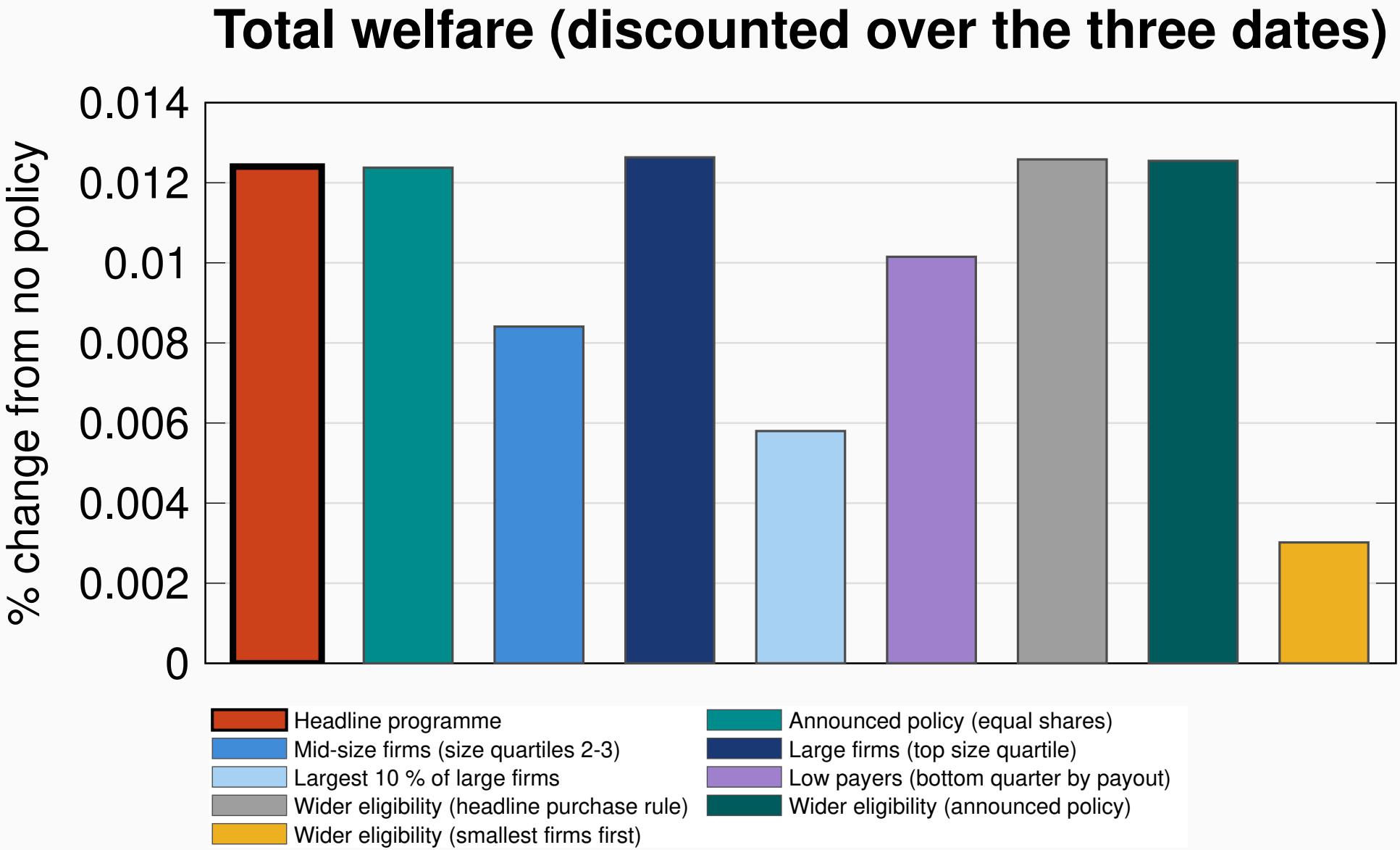
- The realised programme buys **13.14%** of the 70 purchased firms' first issues, which hold 64.9% of eligible issuance: the same total as announced at the first issue $\Rightarrow$ the deviations are a pure redistribution there

- At the repeat issue the realised programme buys $0.823 \times 13.14\%$ of the purchased firms' issues and nothing else: **83%** of the announced repeat-issue total

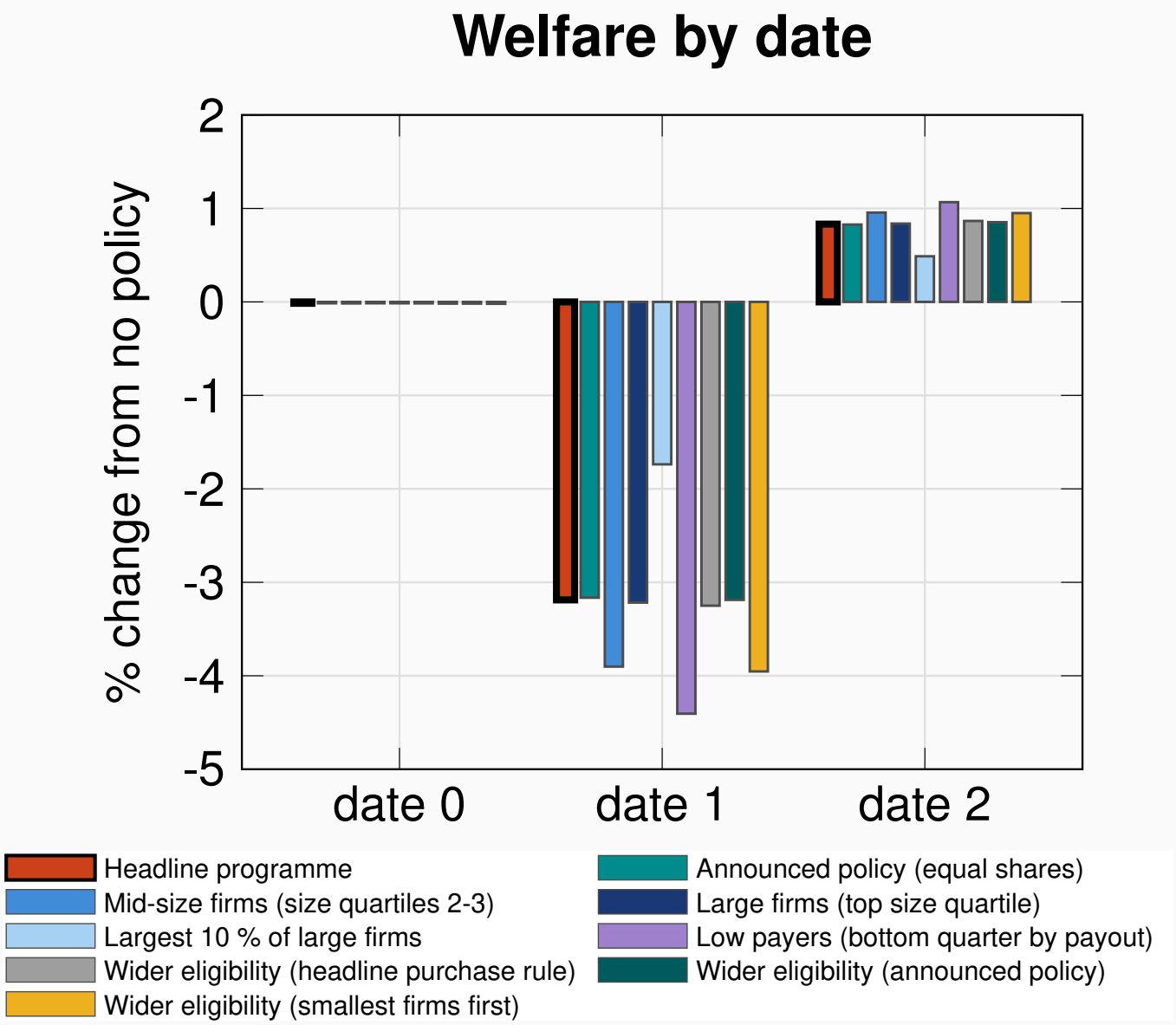
Policy Designs — Markup



Policy Designs — Welfare



Policy Designs — Welfare by Date



Policy Designs — Welfare Decomposition

Back

Design	Extra output	Cost of the extra capital	Private holding cost saved	ECB holding cost	Extra work	Total	Total per euro bought
Headline programme	+0.0232	−0.0114	+0.0039	−0.0008	−0.0024	+0.0124	0.0240
Announced policy (equal shares)	+0.0227	−0.0114	+0.0040	−0.0005	−0.0025	+0.0124	0.0241
Mid-size firms (size quartiles 2–3)	+0.0239	−0.0135	+0.0041	−0.0028	−0.0032	+0.0084	0.0158
Large firms (top size quartile)	+0.0234	−0.0114	+0.0037	−0.0006	−0.0024	+0.0126	0.0243
Largest 10% of large firms	+0.0171	−0.0073	+0.0021	**−0.0047**	−0.0014	+0.0058	0.0108
Low payers (bottom quarter by payout)	**+0.0280**	−0.0147	+0.0034	−0.0033	−0.0033	+0.0102	0.0184
Wider eligibility (headline purchase rule)	+0.0241	−0.0122	+0.0041	−0.0007	−0.0027	+0.0126	0.0242
Wider eligibility (announced policy)	+0.0235	−0.0120	+0.0042	−0.0005	−0.0027	+0.0125	0.0243
Wider eligibility (smallest firms first)	+0.0238	−0.0142	+0.0037	**−0.0068**	−0.0036	+0.0030	0.0052

Central bank cost By date

Welfare With and Without the Central Bank's Holding Cost

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Design	Baseline	No efficiency cost	No central-bank holding cost
Headline programme	0.01240	0.01245	0.01320
Announced policy (equal shares)	0.01238	0.01241	0.01288
Mid-size firms (size quartiles 2–3)	0.00841	0.00857	0.01122
Large firms (top size quartile)	0.01263	0.01267	0.01328
Largest 10% of large firms	0.00580	0.00605	0.01047
Low payers (bottom quarter by payout)	0.01015	0.01033	0.01349
Wider eligibility (headline purchase rule)	0.01258	0.01263	0.01333
Wider eligibility (announced policy)	0.01255	0.01258	0.01302
Wider eligibility (smallest firms first)	0.00302	0.00337	0.00986

Further Purchase Designs

Back Non-payers

Design	Firms bought	Share bought per issue	Programme size (headline = 1)	Capital gained per euro	Output, date 2 (%)	Markup, date 2 (%)	Welfare (%)	Welfare per euro
Headline programme	70	0.131	1.000	0.205	0.437	0.103	0.0124	0.0240
Non-payers (capped at 70% per issue)	6	0.700	0.293	0.278	0.162	0.013	0.0024	0.0160
Non-payers, 5% of the programme size	6	0.134	0.049	0.458	0.048	0.008	0.0017	0.0680
Non-payers, 10% of the programme size	6	0.269	0.102	0.366	0.077	0.010	0.0023	0.0437
Non-payers, 15% of the programme size	6	0.403	0.159	0.343	0.111	0.013	0.0027	0.0333
Non-payers, 20% of the programme size	6	0.537	0.221	0.328	0.145	0.014	0.0029	0.0254
Non-payers, 25% of the programme size	6	0.672	0.281	0.287	0.160	0.014	0.0025	0.0175
Peak (2017) programme, headline frictions	70	0.231	1.790	0.210	0.796	0.185	0.0205	0.0222
Single purchase (no repeat purchase)	70	0.131	0.567	0.026	0.033	0.008	0.0048	0.0163

Purchases of Non-Payers at Different Programme Sizes

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